

Observations on the chemistry, mineralogy and petrology of some Apollo 11 lunar samples

S. O. AGRELL, J. H. SCOON, I. D. MUIR, J. V. P. LONG,

J. D. C. MCCONNELL and A. PECKETT

Department of Mineralogy and Petrology, University of Cambridge, Cambridge, England

(Received 5 February 1970; accepted in revised form 25 February 1970)

Abstract—Chemical analyses, by classical methods, are reported for specimens 10084-14, 10045-24, 10044-39 and 10060-25 along with CIPW norms. The ratio $\frac{(\text{FeO} + \text{MnO}) \times 100}{\text{MgO} + \text{FeO} + \text{MnO}}$ varies from 68 to 74. The analyses confirm the very low silica, the comparatively low alumina and the unusual abundance of titania, (7.75–11.10 per cent). Iron exists almost entirely in the ferrous state and some of the titanium appears to be in the titanous state. Metallic iron figures (0.1–0.6 per cent), obtained by Mössbauer spectra are also reported. Alkali metals are low; varying from 0.36 per cent to 0.49 per cent for Na₂O, and from 0.05 per cent to 0.20 per cent for K₂O.

An account is given of the mineralogy and petrology of fine and of coarse grained igneous rocks and microbreccias with emphasis on the chemical compositions of individual minerals and of glasses. Special attention is paid to the pyroxene relations; both lime-rich and lime-poor magnesian clinopyroxenes including genuine pigeonite are present in the finer rocks. In the later stages of the finer rocks and in the coarser types a trend leading towards subcalcic ferroaugite Ca₂₁Mg₂₁Fe₅₈ is followed. Here the pyroxene grades outwards into a zoned pyroxenoid, pyroxferroite, whose final composition is Ca₁₂Mg₂Fe₈₆. Optical data, including the orientation, are given for this mineral. Pyroxferroite very low in magnesium occurs in the final residuum, associated with ilmenite, baddeleyite, apatite, tridymite and cristobalite, more sodic plagioclase and little patches of a potash feldspar. Electron probe analyses are given of olivines, pyroxenes, pyroxenoids, feldspars, ilmenites, a chrome-titanospinel, armalcolite and a rutile. Analyses of ilmenites (0.97–6.4% MgO and 0.34% Cr₂O₃) and spinel indicate that some of the titanium may be trivalent.

The lunar fines are essentially a sample of the mare surface and the microbreccias are a lithified equivalent. Lithification is in part shock induced and accompanied by vitrification. Both types contain igneous rock and mineral fragments showing a wide range of effects due to shock metamorphism: granulation, maskelynitization, complete fusion and subsequent crystallization. 1–3 mm sized microagglutinate fragments occur; these are coated with vesicular glass with a complex flow pattern shown by metallic iron $\pm$ troilite droplets (5 μm –20 Å). This glass and related glassy spheres have a composition close to the microbreccias and their origin is due to the rapid heating of the surface that accompanied the passage of a hot turbulent cloud of impact-produced particles. Splash glasses were observed and micro-impact craters photographed with the scanning electron microscope. Most of the numerous glassy (3 mm–1 μm spheres) in both microbreccias and fines arose from impact induced melting of small volumes of basaltic rock, or by vapour condensation. An endogenous origin from hot-gas-charged volcanic explosions is not excluded. These glasses are dominantly olivine-normative and may come from a different source or have their compositions changed by differential vaporization. Some spheres are coated, as if sprayed, with metallic iron (<3% Ni) or show a surface concentration of iron (3–19% Ni) and troilite droplets. Electron micrographs reveal craters, burst bubbles, and particulate matter on the exposed surface of the glasses.

Microanorthositic fragments (An 95%) containing geikielo-ilmenite (MgO 9.63%) and rutile occur in soil and microbreccia. Their source is from either a Highland region or a Mare region of anorthite rich basalts; they are the products of rapid crystallization of an impact-produced anorthite-rich melt or the recrystallization product of fine grained anorthite-rich rock. One fragment rich in alkali feldspar (K 77.5 at.%, Na 20.8 at.%, Ca 1.7 at.%) and tridymite was observed, indicating

possible existence of a rock type related to the acid potassic interstitial glass recorded in the igneous rocks. Meteoritic material is rare, <1 per cent: single crystal kamacite and taenite fragments have been observed and one 3 mm granular hexahedrite with kamacite, Ni 6.5% Co 0.5%, and rare zoned droplets (20 μ m) of zoned taenite, core $\sim$ 9% Ni, rim $\sim$ 14% Ni.

INTRODUCTION

THE ROCKS returned by the Apollo 11 mission are all part of the lunar regolith: it may be inferred that some are volcanic rocks originating at no great depth, while others, the microbreccias, form part of a surface layer possibly differing from the dusty mare surface only in degree of compaction. Some components of the regolith show evidence of shock metamorphism and melting. Some of the rock and mineral fragments and most of the glassy spheres may have originated at a considerable distance from the point at which they were collected.

The compositions of individual phases within this material have been determined with a Cambridge Geoscan electron probe operating at an accelerating potential of 20 kV. The experimental procedure and treatment of data used here have been described fully by SWEATMAN and LONG (1969) and all analyses have been referred, either directly or indirectly, to the primary standards listed by these authors. The second decimal place in the percentages obtained is significant at low concentrations and has been retained in all the data quoted. Bulk rock compositions have been determined with classical wet-chemical techniques.

CHEMICAL ANALYSES

The results of chemical analyses by classical methods are set down in Table 1, along with the CIPW norms. Striking features are the low silica values, the comparatively low alumina values and the unusual abundance of titania; iron exists in a very low state of oxidation. Only in analysis 2 is any ferric iron reported. In the three remaining analyses, the ferrous iron estimation is higher than the total iron. This may be explained either by the presence of some of the titanium in the titanous state or by interference in the ferrous iron titration caused by the presence of sulphides. The metallic iron values accepted are based on values obtained by Bancroft from Mössbauer spectra.

An attempt was made to determine the metallic iron content of these samples by subjecting them to attack by a stream of dry chlorine gas (free of HCl) in a tube furnace heated to 350°C, following the method of MOSS and HEY (1961, 1967) for the treatment of meteorites. The volatilized FeCl₃ was then determined by means of a silver reductor, and standard ceric sulphate solution. A correction was made for the presence of FeS which is also attacked under these conditions. These figures appear to be distinctly too high, especially so in analysis 2. The high results are thought to be due to attack on some other mineral phase present, probably an iron-titanium oxide. Tests were made on a Norwegian ilmenite which proved to be completely decomposed by the chlorine treatment. It is unlikely that the ilmenite present in the lunar samples could have been so completely broken down, for much higher results for Fe would have been obtained. It is more likely that a spinel is involved. Investigation of this matter is proceeding.

The SiO₂ was determined by dehydration with HCl and volatilization with HF. Any SiO₂ in solution after this stage was recovered from the pyrosulphate fusion of the ammonia precipitate. The Al₂O₃ was obtained by difference from the ammonia precipitate. Total iron was determined by titration with ceric sulphate solution after treatment in a silver reductor. The FeO was determined

Table 1. Analyses of lunar rocks

	1	2	3	4	5
SiO ₂	42.16	39.60	42.46	41.96	78.92
Al ₂ O ₃	13.60	9.51	10.21	11.85	11.88
Fe ₂ O ₃	nil	0.06	nil	nil	
FeO	15.34	19.12	18.24	16.89	2.88
MnO	0.20	0.28	0.28	0.23	n.d.
MgO	7.76	8.10	5.96	7.63	0.08
CaO	11.94	11.07	12.25	11.38	2.10
Na ₂ O	0.47	0.36	0.48	0.49	0.56
K ₂ O	0.16	0.05	0.11	0.20	6.86
H ₂ O ⁺	0.05	0.15	0.10	0.11	n.d.
H ₂ O ⁻	nil	nil	0.01	0.04	n.d.
TiO ₂	7.75	11.10	9.18	9.02	n.d.
P ₂ O ₅	0.05	0.04	0.04	0.07	n.d.
Cr ₂ O ₃	0.30	0.47	0.21	0.31	n.d.
S	0.12	0.14	0.18	0.15	n.d.
F	nil	nil	nil	nil	n.d.
Fe	0.60	0.20	0.10	0.30	—
less S=O	100.50	100.25	99.81	100.63	103.28
	0.03	0.04	0.05	0.04	
Totals	100.47	100.21	99.76	100.59	
Total					
Fe as					
Fe ₂ O ₃	17.56	21.59	20.85	18.90	
CIPW norms					
Q	0.24	0.81	3.64	1.36	41.2
F {	Or 0.95	0.30	0.65	1.18	39.3
	Ab 3.98	3.05	4.06	4.15	4.6
	An 34.53	24.19	25.38	29.55	9.3
Di {	Wo 10.18	12.72	14.67	11.05	
	En 5.53	6.82	6.60	5.87	0.7 (Di)
	Fs 4.29	5.49	7.98	4.84	
Hy {	En 13.79	13.36	8.24	13.14	5.0 (Hy)
	Fs 10.70	10.76	9.96	10.84	
	Cr 0.44	0.69	0.31	0.46	
	Ilm 14.72	21.08	17.44	17.13	
	Mt nil	0.09	nil	nil	
	Ap 0.12	0.09	0.09	0.17	
	Tr 0.32	0.38	0.49	0.41	
	Fe 0.60	0.20	0.10	0.30	
Totals	100.39	100.03	99.62	100.42	100.0

1. Apollo 11. 10084-14. Fines: Fe 1.22 by CI method.
2. Apollo 11. 10045-24. Type A: Fe 1.78 by CI method.
3. Apollo 11. 10044-39. Type B: Fe 1.23 by CI method.
4. Apollo 11. 10060-25. Type C: Fe 1.18 by CI method.
5. Apollo 11. 10017-50. Type B: electron-probe analysis of interstitial glass. The high total is due to loss of alkalis during measurement of silica.

Cr chromite; Tr troilite; Fe metallic iron.

by the modified Pratt method. The MnO was obtained colorimetrically after oxidation with KIO₄. The MgO was precipitated as magnesium ammonium phosphate and ignited to Mg₂P₂O₇. The CaO was precipitated as oxalate and ignited to oxide. The Na₂O and K₂O were determined using an E.E.L. flame photometer.

The H₂O⁻ was obtained from the loss in weight at 105°C. The TiO₂ was estimated colorimetrically using H₂O₂. The P₂O₅ was determined colorimetrically by the vanadomolybdate method. The Cr₂O₃ was determined by the diphenylcarbazide method. Sulphur was precipitated as BaSO₄ after

Na_2CO_3 fusion and extraction of the cake with water, allowance being made for reagent blank. Fluorine was determined by the Sen Gupta modification of the method of HUANG and JOHNS (SEN GUPTA, 1968).

The values obtained for total water appear to be higher than those obtained by other investigators; a more detailed account of the method used is therefore set out below.

About 0.5 g of PbCrO_4 was introduced by means of a thistle funnel to a clean, dry Penfield tube made of Pyrex glass, 25 cm long and of 1 cm dia. The PbCrO_4 was heated in a Bunsen flame until it just commenced to melt. The upper part of the tube was heated and any water evolved was sucked out by means of a stream of air from a vacuum pump. The tube was then stoppered and allowed to cool thoroughly. About 0.3–0.4 g of the lunar sample was then weighed into the tube, and the water expelled by heating strongly in an oxygen–natural gas blowpipe flame. The remainder of the determination was carried out according to the normal Penfield procedure. A similar determination was carried out on the PbCrO_4 alone, and no blank was obtained. Subsequent investigation of this matter, however, indicates that this blank might be as high as 0.0003 g when 0.5 g of flux is used, thus throwing some doubt on the values of H_2O^+ shown in the table. The flux was necessary in order to retain sulphur which might otherwise be evolved with the water.

MINERALOGY AND PETROLOGY: IGNEOUS ROCKS

Thin sections were examined from four igneous rocks: 10003-44 (Type B); 100017-50 (Type B but transitional to Type A); 10045-35 (Type A); and 10058-23 (Type B), probably the coarsest grained rock of the entire Apollo 11 suite. In addition ferromagnesian mineral fractions from 10044-43 (Type A) were examined in liquid mounts, and small fragments, some hardly altered, some thoroughly recrystallized, were studied in the breccia 10065-15. Chemical analyses of 10044-43 and of 10045-34 are reported in this paper by SCOON (Table 1), of 10003 and 10058 by ROSE *et al.* (1970), and of 10017-20 by WILK (1970). Trace element data on 10044-39 and 10045-24 are given by BROWN *et al.* (1970).

The optical properties of the pyroxenes, pyroxenoid, amphibole and plagioclase were investigated with a Leitz 4-axis Universal stage. Conoscopic methods were employed, using a small upper hemisphere and objective $\times 32$ to determine the homogeneous areas of the pyroxenes and pyroxenoid in 10058 and the crystals of amphibole derived from this rock; the position of the melatope was located within 1° by using two matched $\frac{1}{4}\lambda$ plates. In the finer grained rocks and in the outer more strongly zoned parts of the pyroxene–pyroxenoid grains, orthoscopic methods were employed using standard radius hemispheres $n = 1.760$, as conoscopic methods cannot be applied to material in which there is a rapid variation in the optical properties. Standard Fedorov procedures were employed to study the plagioclase but these were accompanied by a determination of the maximum extinction angle in the (010) zone by the Rittmann method; this proved useful in enabling a rapid survey to be made of the range of zoning in each specimen and of the homogeneity of the larger crystals. The optical properties of the amphibole crystal used for the X-ray investigation (Gay *et al.*, 1970) were obtained by mounting a fragment on a fibre and securing it to the spindle of the attachment described by WILCOX (1959) for use with the Universal stage; the (010) plane was oriented normal to the spindle axis. A single variation procedure was used to determine the refractive indices.

Petrography

The grain size of the igneous types examined ranges from that typical of a medium grained basalt to that of the coarsest dolerite; the Type B rocks are holocrystalline; there is a little potassium-rich acid glass in the residuum of 10017 (Table 1, analysis 5), and rather more in 10045. All four rocks are similar in mineralogy and all are quite different from any known terrestrial type by virtue of their very high titania content

and their very reduced state. If total iron is regarded as FeO, the values of the ratio $\frac{(\text{FeO} + \text{MnO}) \times 100}{\text{MgO} + \text{FeO} + \text{MnO}}$ vary from 67 to 75 for the analysed rocks. No terrestrial types with similar ratios are so poor in soda. At this stage it would seem prudent to defer the naming of the rocks until it is known how unusual for lunar maria rocks is the very high titania content of the Apollo 11 suite. In this paper the terms basalt and dolerite will be applied merely in a textural capacity.

Three minerals, clinopyroxene, bytownite and ilmenite (in decreasing order of abundance) are prominent phases in all of the rocks and of these clinopyroxene predominates absolutely. In 10045 there are up to five per cent of phenocrysts and microphenocrysts of olivine many of which are enclosed by clinopyroxene; some are sharply jacketed by it (Fig. 1) suggesting a reaction relation similar to that seen occasionally in terrestrial basanitoids. In their pyroxene relations the rocks bear a superficial resemblance to tholeiites or to eucrite meteorites, the latter carrying similar plagioclase (Juvinas, An_{84}). In 10003, 10045 and 10058 the pyroxene grades outwards into, or is mantled by, an iron-rich pyroxenoid which has continued to crystallize until a late stage. The pyroxenoid was not observed in the single thin section of 10017 that was available. It is very unevenly distributed and is the dominant phase in the single small section of 10003 that was examined. In 10058 coarse patches of low cristobalite and of low tridymite are found in the residuum. Some areas with the mosaic texture of cristobalite have a plate-like habit and may be paramorphs after tridymite. These silica minerals are sometimes intergrown with a more sodic plagioclase or with little patches of alkali feldspar. Late ilmenite, a spinel close to ulvöspinel, and baddeleyite occur in or adjacent to the residuum of 10058. Hedenbergite and fayalite were sought but not found.

Accessories are a brown spinel (picotite) occurring as tiny octahedra enclosed in the olivine as in alkali basalts, and armalcolite of early crystallization that is mantled by ilmenite has been described by SIMPSON and BOWIE (1970). A few needles of a brown mineral thought to be rutile occur in the residuum with slender prisms of apatite. Other accessories are troilite and an occasional speck of a sulphide phase resembling chalcopyrite. Closely associated with the troilite is metallic iron either as small droplike areas 1–5 μm in dia. or as grains in which crystal form is weakly developed. The relationship of these sulphide-metal areas to the surrounding silicates suggests an immiscible relationship between the later stage silicate liquid and a sulphide-metal liquid (see also SIMPSON and BOWIE, 1970).

In the heavy residue of 10044 one grain of an alloy of composition approximately $\text{Cu}_{70}\text{Zn}_{30}$ was found. As this composition is in the range of commercial brass, contamination was suspected but an additional grain of the same material has been found and this had a troilite and calcic plagioclase crystal firmly attached to it (GAY *et al.*, 1970). In rock 10017-50 a firmly attached crystal was observed. This was removed and its composition is reported on by GAY *et al.* (1970).

From small vugs in 10058, crystals of calcic plagioclase, a blue amphibole, tridymite, and ilmenite were found. Aragonite was found in the container that held the specimen and may have fallen out of a vug. The presence of these minerals is important in indicating the role of a vapour phase in the crystallization process. If

contamination can be excluded, it seems likely that other minerals such as mica and calcite (AGRELL, 1970) reported from the lunar soil may be derived from this source.

Textures range from subpoikilitic in 10003 and 10058 where the grain size of the ferromagnesian minerals much exceeds that of plagioclase, to intergranular in 10017 and 10045 where lath shaped plagioclases separate areas composed mainly of small overlapping equant pyroxene grains, pyroxenoid and ilmenite. In these latter rocks the grain size of the plagioclase greatly exceeds that of the ferromagnesian silicates and plagioclase laths in 10017 may approach 3 mm in length. The texture of this rock is similar in some ways to that of the perpendicular feldspar rock from the Skaergaard intrusion, a marginal facies in which the large subhedral plagioclases are considered to have grown rapidly under tranquil conditions of considerable undercooling. In both 10017 and 10058 Universal stage studies suggest some degree of preferred orientation in the plagioclase. It has been suggested (CHAO *et al.*, 1970) that the textures of 10017 are developed as a result of metamorphism which has led to the recrystallization of the pyroxenes as small interlocking grains, but the presence of very strong zoning marginally within the confines of single small crystals (as shown by the rapid variation in $2V$, and the marginal zoning of the plagioclase) argues against this.

The textural features are consistent with the finer rocks being derived from lava flows or from the chilled margins of minor intrusions; the coarser types may represent the interiors of flows but are more likely to be derived from minor intrusions such as sills or inclined sheets located at no great depth. This interpretation is consistent with the development of cristobalite and tridymite, the mild degree of vesiculation, the presence and nature of the vug minerals, and the evidence for small amounts of post-crystallization reactions observed in the late stage minerals. It is conceivable that sodium, and possibly potassium, along with silica, may have been lost during the escape of the vapour phase. Flow structures have not been observed nor is there any evidence of the formation of cumulates.

Mineralogy

(i) *Olivine*

Olivine commenced and ceased to crystallize at an early stage; it is present in amounts of up to 5 per cent in 10045 and contains a few octahedra of coffee-brown picotite. Crystals seldom exceed 0.5 mm in length, display evidence of slight zoning, and are frequently enclosed in clinopyroxene. In all cases where the nature of the surrounding pyroxene could be determined this proved to be a normal augite with $2V$ close to 40° . Probe analyses of the olivines from this rock (not presented here) indicate compositions in the range Fa_{25-30} which confirm the estimates based on $2V$ which indicate slight zoning in the range $2V_x$ 83° – 81° (Fa_{26-32}). Olivine has also been reported in 10017 (REED *et al.*, 1970). Glide lamellae have not been observed.

Analyses of olivines from the fines (10085) and from the microbreccia (10065) are presented in Table 2. They cover approximately the same composition range as do those from 10045. Significant features are the consistent values for lime and titania, the very low values of nickel, and high chromium in analyses 1 and 4. In commenting on the consistently high chromium values found on the Apollo 11 olivines, HAGGERTY

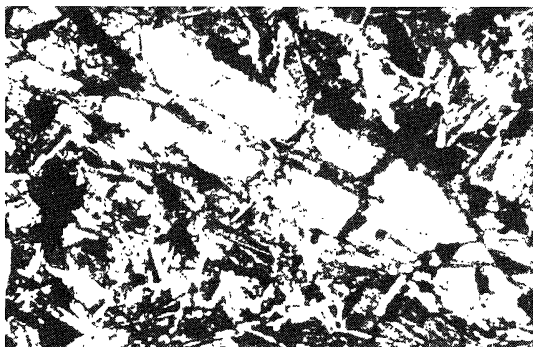


Fig. 1. 10045-34. Crystal of olivine with a swallow-tail-termination (upper left), and a narrow reaction border of clinopyroxene. The groundmass is composed mainly of granules of pyroxene, some pyroxenoid, laths of calcic plagioclase and plates of ilmenite. Width of field 3.5 mm.



Fig. 2. 10058-23. Enlargement of lower left of crystal of clinopyroxene-pyroxenoid shown in Fig. 3 to show a rim of turbid magnesia-poor pyroxenoid where it is in contact with very late ilmenite or the residuum. Residuum here is composed of low cristobalite, tridymite (twinned crystal extreme upper right), more sodic plagioclase (multiple twinning), a little alkali feldspar, and opaques. Crossed polars. Width of field 1.5 mm.



Fig. 3. 10058-23. Strongly zoned crystal of clinopyroxene-pyroxenoid described in the text from which analyses 3-9 of Table 4 were obtained. Note the continuous variation in the birefringence. Crossed polars. Width of field 3.5 mm.



Fig. 4. 10003-44. Pyroxenoid: note the tendency to develop a mosaic texture, non-uniform extinction, and the tendency towards curvature of the cleavage planes. Crossed polars. Width of field 3.5 mm.

Table 2. Olivine analyses

	1	2	3	4		1	2	3	4
						Number of ions on basis 0 = 4			
SiO ₂	37.53	37.74	38.32	37.85	Si	0.992	0.996	1.000	0.988
TiO ₂	0.16	0.19	0.12	0.34	Ti	0.003	0.004	0.002	0.007
Al ₂ O ₃	n.d.	0.16	0.19	0.30	Al	—	0.005	0.006	0.009
Cr ₂ O ₃	0.27	0.06	0.06	0.22	Cr	0.006	0.002	0.002	0.005
FeO	25.46	25.79	22.55	22.00	Fe	0.563	0.569	0.492	0.480
MnO	0.27	0.33	0.27	0.23	Mn	0.006	0.007	0.006	0.005
MgO	36.08	35.72	37.96	38.45	Mg	1.422	1.406	1.476	1.495
CaO	0.32	0.28	0.37	0.34	Ca	0.009	0.008	0.010	0.010
NiO	0.03	n.d.	n.d.	0.05	Ni	0.001	—	—	0.001
					(Mg/Mg + Fe + Mn) × 100	71.4	70.9	74.8	75.5
Totals	100.12	100.27	99.84	99.78					

1. 10085-4-5. Phenocrysts in brown glass, thick coating type, in fines.

2, 3. 10085-4-10a. Drop-like crystals in microanorthosite fragment in fines.

4. 10065-15-1a. Phenocryst in quenched oxide-rich basalt, fragment in microbreccia.

et al. (1970) pointed out that spectrographic evidence suggested that some or all of this element was in the divalent state.

Fayalite was sought but not found in 10058. The late-stage turbid mineral mantling the pyroxenoid adjacent to ilmenite in the residuum (Fig. 2) proved to be an almost magnesia-free pyroxenoid. The presence of fayalite in this environment as marginal outgrowths to and as a possible replacement of the pyroxenoid, has been reported by REED *et al.* (1970). Fayalite has been reported in the residuum as tiny grains or needles forming out-growths to hedenbergite (see also SKINNER, 1970; ANDERSON *et al.*, 1970). As yet no analyses of this fayalite are available.

(ii) *Pyroxene*

In general, a single clinopyroxene phase is present which in the coarse dolerites displays remarkably little variation in $2V$ in spite of the extensive range of chemical zoning that prevails within a single crystal. Zoning may take a number of forms; in the coarsest rocks the hourglass variety so typical of titanaugites may be observed; in this type large single crystals may be composed of a number of slightly disoriented units, or of sectors. Outwards an irregular type of planar zoning may be superimposed but planes of physical discontinuity, possibly low-angle planes produced by the pile-up of dislocations, often with a steplike character, may occur as well, and there is some variation in the birefringence. Some crystals display a few broad glide lamellae parallel to (001). In 10017 and 10045 the early augite contains little patches of pigeonite [optic plane $\perp$ (010)] but this mineral has not been found in the coarser types that have been examined. A remarkable feature in the dolerites is the abundance of subcalcic augite. The clinopyroxene of these rocks is zoned from cores $2V = 48^\circ$, $Z \wedge c = 37^\circ$, and with moderate dispersion of optic axis A , (3°), to margins $2V = 38^\circ$, $Z \wedge c = 40^\circ$ and with less marked dispersion. The range of n_β for this pyroxene extends approximately from 1.700 to 1.755. This pyroxene grades outwards into or is sharply mantled by the pyroxenoid. Probe analyses indicate zoning from $\text{Ca}_{35}\text{Mg}_{45}\text{Fe}_{20}$ to a subcalcic ferroaugite near $\text{Ca}_{21}\text{Mg}_{21}\text{Fe}_{58}$, almost the complete range being exhibited within a

single crystal (Fig. 3). The complete crystallization trend shown for 10058 in Fig. 5 is derived from point analyses within the crystal illustrated in Fig. 3. Analyses of other crystals indicate a similar trend but with some variation in the lime content. Some are very similar to those reported by BROWN *et al.* (1970) in which, although the compositions are more calcic, the trend is similar. However the variations between one crystal and another in this rock are no greater than might be reasonably attributed to scatter in a rock of this type. Similar variations have been encountered within specimens of dolerites from the upper part of the Mt. Wellington sill, Tasmania (MUIR, unpublished data).

In the finer rocks 10017 and 10045 a very different trend emerges which is characterized by a very large variation in $2V$. The early pyroxene, $2V = 48^\circ$, $\text{Ca}_{44}\text{Mg}_{41}\text{Fe}_{15}$ is richer in lime and poorer in iron than the earliest pyroxene of the dolerite and in some crystals it can be shown to be zoned continuously from $2V\ 42^\circ$ to 20° with the optic plane remaining parallel to (010), although the complete range is not seen in a single crystal. Irregular zoning may also occur with occasional patches of low $2V$, (about 25°) being found at the sides or centres of small grains in a similar manner to those described by MUIR and LONG (1965) in Mauna Loa basalts. But the predominant trend is towards a lower $2V$ value and a larger extinction angle as the margins of grains are approached. No late pyroxene with the optic plane $\perp$ (010) was observed. Although in a rock such as 10045 a considerable scatter in the composition of the groundmass pyroxene would be expected, two distinct trends are revealed. The first is shown by the probe analyses of Table 3 which plot almost in a line on Fig. 5; this terminates in a pigeonite $\text{Ca}_{15}\text{Mg}_{54}\text{Fe}_{31}$. The second trend indicating strong iron enrichment in the subcalcic augite series is revealed by the probe analyses reported by BROWN *et al.* (1970). This must indicate a strong fractionation in the late liquids which culminates in the formation of the pyroxenoid mineral. The moderately magnesia-rich overall composition of the pyroxene and pyroxenoid phases in this rock is revealed by the normative pyroxene plotted in Fig. 5. The analyses in Table 3, and the clear optical evidence of zoning might be taken to imply that in this rock the principal course of pyroxene fractionation is following a distinctly magnesia-rich variant of the tholeiitic quench trend (MUIR and TILLEY, 1964). An alternative interpretation would be that the early pyroxenes display a remarkable field of metastable scatter which is constrained in the later stages by the failure of iron-rich pigeonites to crystallize. Presumably hypersolvus crystallization is involved at this stage. A similar trend, but with a slightly higher degree of iron enrichment in the lime-poor pyroxenes appears in the slightly coarser 10017, although the rock itself (as shown by its normative pyroxene) is more magnesian. The complete absence of iron-rich pigeonites as a separate phase is a striking feature in these lunar rocks; it is brought out well in the diagram given by ARRHENIUS *et al.* (1970) where the overwhelming concentrations of plotted points define the trends revealed by the analyses of 10045 and 10058. Nevertheless it is clear that at a moderately late stage pyroxenes ceased to crystallize and their place was taken by the pyroxenoid mineral. Only in the latest stages of crystallization of the mesostasis, do they reappear, associated with silica minerals, and then the composition is close to hedenbergite. At this stage the pyroxenoid is unstable.

Table 3. Pyroxene analyses in 10045-35 and 10085-4-2

	1	2	3	4	5	6	7	8
SiO ₂	48.00	49.96	52.06	50.78	49.09	49.71	49.35	52.18
TiO ₂	5.91	3.29	1.35	2.03	3.30	1.59	1.94	0.83
Al ₂ O ₃	4.93	2.83	0.85	1.92	3.49	1.66	2.39	1.00
Cr ₂ O ₃	0.70	n.d.	0.25	n.d.	n.d.	n.d.	n.d.	n.d.
FeO	8.49	15.90	19.15	10.59	8.71	20.55	13.46	18.01
MnO	0.09	0.29	0.35	0.28	0.22	0.44	0.28	0.37
MgO	13.27	13.93	18.61	16.35	13.80	13.05	12.82	20.55
CaO	19.88	14.19	6.89	16.47	20.10	12.26	17.82	5.24
Totals	101.27	100.39	99.51	98.42	98.71	99.26	98.06	98.18
Number of ions on basis 0 = 6								
Si	1.770	1.881	1.960	1.915	1.854	1.926	1.904	1.969
Al	0.214	0.119	0.030	0.085	0.146	0.074	0.096	0.081
Al	—	0.007	—	—	0.009	0.002	0.013	0.013
Cr	0.020	—	0.007	—	—	—	—	—
Ti	0.164	0.093	0.038	0.058	0.094	0.046	0.056	0.024
Fe	0.262	0.501	0.603	0.334	0.275	0.666	0.434	0.568
Mn	0.003	0.009	0.011	0.009	0.007	0.014	0.009	0.012
Mg	0.729	0.782	1.044	0.919	0.777	0.753	0.737	1.156
Ca	0.786	0.572	0.278	0.665	0.813	0.509	0.737	0.212
Atomic %								
Ca	44.2	30.7	14.4	34.5	43.4	26.2	38.4	10.9
Mg	41.0	42.0	53.9	47.7	41.5	38.8	38.5	59.2
Fe+Mn	14.8	27.3	31.7	17.8	15.1	35.0	23.1	29.8
1. 10045-35. Augite (Ti-rich) 2. 10045-35. Subcalcic augite 3. 10045-35. Pigeonite } from basaltic rock 4. 10085-4-2. Paler augite crystal 5. 10085-4-2. Darker augite crystal 6. 10085-4-2. Sub-calcic augite 7. 10085-4-2. Augite 8. 10085-4-2. Pigeonite } composite crystal group } from coarse basaltic rock in fines.								

Analyses of pyroxenes from 10045 and from the fines 10085 are given in Table 3 and of those from 10058 in Table 4, analyses 1–5. They reveal that the earliest pyroxene (Table 3, analysis 1) is significantly aluminous (nearly 5 per cent) and contains nearly 6 per cent titania; except for the absence of ferric oxide this and the earliest pyroxene from 10058 are genuine titanaugites. Notable also are the significant chromium contents, although this constituent is higher in the augite of 10045 than it is in the pigeonite. As fractionation proceeds there is a striking reduction in the values of both alumina and titania but in 10058 although chromium also decreases it remains significant even in the subcalcic ferroaugite, a feature that is in marked contrast to that of terrestrial pyroxenes when there is any significant iron enrichment.

Exsolution lamellae were not observed optically in even the coarsest rocks but X-ray studies (GAY *et al.*, 1970) indicate that they are present on a submicroscopic scale. A possible explanation of the absence of visible lamellae in such coarse grained rocks and of the abundance of subcalcic augite must lie in some factor depressing the pyroxene solvus or rendering it less effective. It was earlier suggested that the titania (AGRELL *et al.*, 1970) contents of the pyroxene might achieve this but this now seems unlikely in view of the evidence for the reduction in the titania content of the pyroxene as fractionation proceeds. It seems more likely that the dry state of the magma may

Observations on the chemistry, mineralogy and petrology of some Apollo 11 lunar samples 103

Table 4. Analyses of pyroxenes and pyroxenoids from 10058-23 and 10065-15

	1	2	3	4	5	6	7	8	9	10
SiO ₂	49.32	51.23	50.36	49.76	48.31	46.92	46.31	45.92	45.86	46.41
TiO ₂	2.49	1.46	1.67	1.30	0.66	0.74	0.63	0.63	0.59	1.13
Al ₂ O ₃	3.17	1.69	1.97	1.49	0.81	0.76	0.64	0.88	0.74	1.16
Cr ₂ O ₃	0.45	0.24	0.31	0.24	0.14	0.12	0.10	0.09	0.10	0.13
FeO	10.95	14.49	17.03	22.97	36.11	42.48	42.45	42.45	44.19	37.88
MnO	0.08	0.26	0.32	0.43	0.65	0.65	0.64	0.62	0.64	0.61
MgO	14.80	16.27	14.17	11.69	5.90	2.41	1.63	1.59	1.38	2.14
CaO	17.95	13.98	13.85	12.07	8.97	6.69	6.37	6.48	6.21	10.21
Na ₂ O	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	0.08
Totals	99.21	99.62	99.68	99.95	101.55	100.77	98.77	98.66	99.71	99.75
Number of ions on basis 0 = 6										
Si	1.861	1.927	1.918	1.935	1.954	1.965	1.983	1.971	1.962	1.946
Al	0.139	0.073	0.088	0.065	0.039	0.035	0.017	0.029	0.037	0.054
Al	0.003	0.002	—	0.002	—	0.003	0.015	0.016	—	0.003
Ti	0.071	0.041	0.048	0.038	0.020	0.023	0.020	0.020	0.019	0.036
Cr	0.013	0.007	0.009	0.007	0.004	0.004	0.003	0.003	0.003	0.004
Fe	0.345	0.456	0.542	0.747	1.221	1.488	1.520	1.524	1.581	1.328
Mn	0.003	0.008	0.010	0.014	0.022	0.023	0.023	0.023	0.023	0.022
Mg	0.832	0.912	0.804	0.677	0.356	0.150	0.104	0.102	0.102	0.134
Ca	0.726	0.564	0.565	0.503	0.389	0.300	0.292	0.298	0.285	0.459
Na	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	0.007
Atomic %										
Ca	38.1	29.1	29.4	25.9	19.6	15.3	15.1	15.3	14.4	23.6
Mg	43.7	47.0	41.9	34.9	17.9	7.6	5.4	5.2	4.5	6.9
Fe + Mn	18.2	23.9	28.7	39.2	62.5	77.1	79.6	79.5	81.1	69.5

1. 10058-23. Pyroxene.
2. 10058-23. Pyroxene.
- 3-9. 10058-23. Analyses from core, 3, to rim, 9, of an apparently continuously zoned crystal. 3-5 pyroxene, 6-9 pyroxenoid. Analyses 1-9 from coarse doleritic rock.
10. 10065-15. Pyroxenoid fragment in microbreccia.

be responsible. Orthopyroxene has not been found in any of the igneous types examined but fragments have been reported in some of the breccias. It may occur in one very fine grained granulitic fragment in 10065-15 where the original pyroxene has recrystallized as two separate phases, one of which has the characteristic birefringence of magnesian orthopyroxene.

(iii) *Pyroxenoid**

This mineral, which is a distinct yellow when viewed through the stereo-microscope displays a very pale yellow colour in thin section which becomes somewhat more distinct towards the margins; but the colour cannot be observed if even a very pale blue filter is employed. In thin section it tends to split parallel to the cleavages which imparts a fibrous appearance; there is a tendency also towards curvature. This feature is particularly well shown in 10003 (Fig. 4) where the mosaic texture of interlocking

* The existence of this mineral was made known to us by J. V. SMITH (personal communication) in November 1969; he suggested the name pyroxferroite and pointed out its relationship to pyroxmangite.

grains is also particularly well displayed. This latter character, and the tendency towards undulose extinction are shown also in monochromatic light. It is even shown sometimes when there is no curvature. These features are reminiscent of those displayed by the green pyroxenes in the Skaergaard intrusion that have formed by inversion from iron wollastonite.

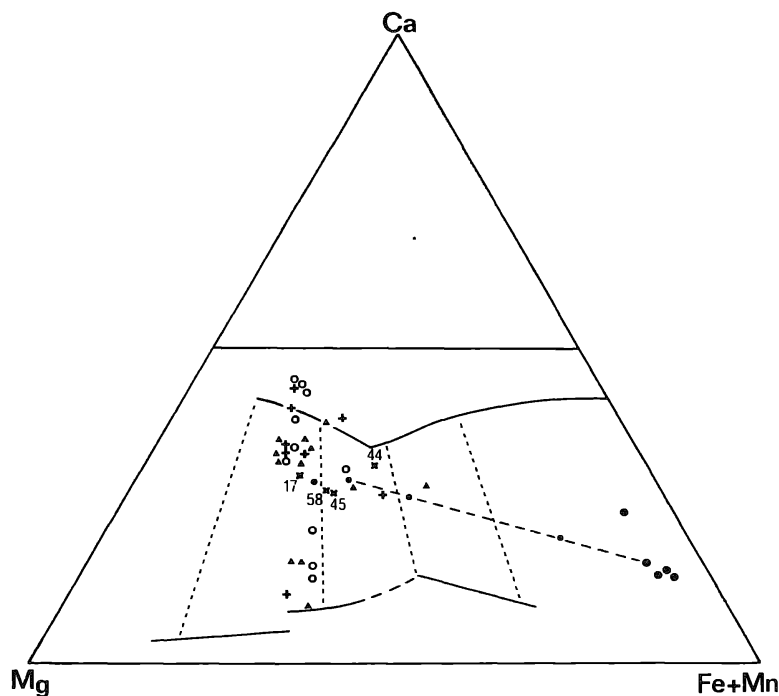


Fig. 5. Compositional variation in the clinopyroxene and pyroxenoid of some lunar rocks in terms of atomic per cent Ca, Mg, Fe + Mn.

- 10045-35 Fine-grained basaltic rock.
- + 10085-4-2 Medium-grained basaltic rock, fragment from fines.
- ▲ 10017-12 Medium-grained basaltic rock.
- 10058-23 Fine-grained gabbroic rock.
- Apparently continuous zoning in a single crystal with a core of pyroxene “●” and a rim of pyroxenoid “●”.
- 10058-23, 10060-15 Pyroxenoid
- × Normative pyroxene from rocks 10044, 10045 (Table 1) and 10017, 10058.

The trend of lime-rich and lime-poor pyroxenes of the Skaergaard intrusion is indicated.

The lime-poor trend is taken from unpublished data by I. D. Muir.

The pyroxenoid occurs intergrown with or mantling the clinopyroxene and as separate small grains of late crystallization, which continue into the final residuum. In some cases it sharply mantles the clinopyroxene and the boundary is displayed clearly by a change in the birefringence; in other cases there appears to be a gradual change with no compositional break but with a gradual increase in the triclinicity and partial birefringence as the pyroxenoid takes over and grades outwards towards its still more iron-rich margins. This effect is beautifully illustrated by the composite grain shown in Fig. 3 which shows a first order orange in the homogeneous pyroxene center of the grain and this grades outwards to red. In the pyroxenoid part of the grain,

in the dark region of the crystal in the figure, there is a rapid change to second order blue. In the orientation shown, the wave normal is approximately equally inclined to all three optic symmetry planes. Points represented by the analyses given in Table 4 are indicated on the crystal, and subsequent partial analyses for Ca, Mg and Fe indicate no break in composition. The appearance of a gradual change could be produced if a plane corresponding with a break in composition was nearly parallel to the plane of the section, but when this crystal was inclined on the Universal stage no sharp boundary was detected. The continuous nature of the change in chemical composition from zoned clinopyroxene to zoned pyroxenoid is revealed by similar marked changes in the birefringence displayed by more than half a dozen crystals in the same thin section. Confirmation has been obtained by subsequent probe investigations of a further two. Towards the extreme margins there is a very rapid change in the composition of the pyroxenoid. In the crystal illustrated in Fig. 3 a slight but irregular feature indicating some physical discontinuity associated with a minor refractive index difference was observed when the section was inclined by about 35° , but this discontinuity is less obvious than are those associated with the pile-up of dislocations or with zoning in the parts of the crystal that are unquestionably clinopyroxene.

An attempt was made with the Universal stage to establish in this crystal the exact position where triclinic optical properties begin. The (010) plane can be identified from the optics of the clinopyroxene portion just beyond the glide lamellae; this was oriented conoscopically. In the outer part of the crystal the coplanar extinction curve method (orthoscopic) of JOEL and MUIR (1958) was employed using standard radius hemispheres $n = 1.760$. Although this showed that the optic plane was rotated by about 2° just inwards from the point represented by analysis 5, Table 4, unfortunately at this point there is a slight disorientation in the crystal revealed by the setting of the cleavage. $2V$ in the pyroxenoid is 38° here and the triclinicity and the extinction angle on (010) appear to increase outwards towards the margin. In another crystal initially oriented nearly normal to [001] a slight change in the cleavage angle, the appearance of the broken-up character, and the presence of poorer third and fourth cleavages suggest a change in the nature of the mineral but this can only be discerned when the cleavages are nearly vertical.

In this crystal, which is distinctly more calcic than the one illustrated, the change in cleavage character occurs at $2V = 38^\circ$. At this stage, the mineral is just triclinic optically and both the small extinction angle, 1° , to the (010) cleavage in a section normal to [001], and the extinction angle on (010) increase towards the margin where $Z \wedge c = 47^\circ$ and $2V = 31^\circ$. Subsequent investigation of this grain with the probe indicated the surprising composition of $\text{Ca}_{27}\text{Mg}_{33}\text{Fe}_{40}$ where the change to triclinic optics is found. As there is evidence of severe fracturing and plucking in this region it is possible that strain due to polishing may be the explanation. Tentatively, the boundary between clinopyroxene and pyroxenoid has been placed at a composition slightly less magnesian than the point indicated by analysis 5 of Table 4.

The complete optical orientation was established by conoscopic measurements on a small (100) twinned crystal wedged in between the larger pyroxene–pyroxenoid grains which displayed fairly uniform extinction. The stereogram showing the orientation, the position of the cleavages and their relation to that of the pyroxene in this rock, is

given in Fig. 6. This demonstrates clearly the triclinic character. The optic plane does not bisect the angle subtended by the good cleavages and makes an angle of between 2.5 and 4° with the poorer (010) cleavage. The fourth prismatic cleavage is very poor; it subtends an angle of 20° – 25° with (110) and by analogy with the others appears to be (120). The acute angle between the cleavages seems to vary between $82\frac{1}{2}^\circ$ and 87° in different parts of the mosaic mantling region. Here indices have been assigned to the cleavages by analogy with those of the closely related pyroxenes. There may also be a (001) parting. No trace of a ($\bar{1}20$) cleavage could be detected. Except for (120), these cleavages are the same as those reported for pyroxmangite.

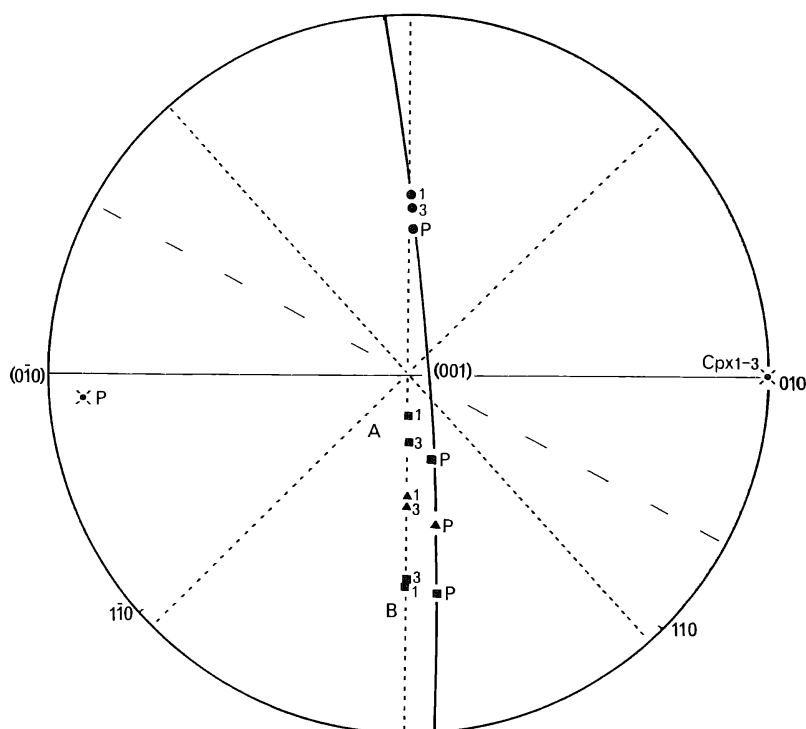


Fig. 6. Stereogram showing the optical orientation of the pyroxenoid, P, and its relation to that of the zoned pyroxene it mantles. 1 early pyroxene, 3 late pyroxene. $\bullet$ X, $\times$ Y, $\blacktriangle$ Z, $\blacksquare$ optic axes A and B. Cleavages are indicated by dashed lines; wider spaced dashes, fourth cleavage (120).

Twins on (100) are as common as in clinopyroxenes and in such cases the optic axial planes of the two subindividuals intersect at an angle of 5° : $2V_z$ usually lies in the range 38° – 34° and is 31° for the most lime-poor and iron-rich zone. In the more lime-rich pyroxenoid from the breccia 10065 (Table 4, analysis 10), $2V = 42^\circ$. The higher values of $2V$ previously reported (AGRELL *et al.*, 1970) were made on regions of zoned pyroxene–pyroxenoid grains now known to be clinopyroxene. In the pyroxenoid $Z \wedge c = 42^\circ$ – 45° and up to 47° at the extreme margins; there is little dispersion of the optic plane but considerable dispersion of the XY and YZ planes: pleochroism is not obvious in thin section. Refractive indices determined on grains from 10044-43 are $n_\alpha = 1.749$ – 1.753 ; $n_\beta = 1.753$ – 1.756 ; $n_\gamma = 1.765$ – 1.768 : all ± 0.001 . Grains from 10058 were not available and time has not permitted removals from thin sections,

but an extended range of zoning is likely. Approximate determinations of the birefringence in zoned grains with the Berek compensator indicated an increase in the birefringence from 0.016 to 0.024 outwards. This was based on an estimate of the thickness made by assuming the birefringence of the clinopyroxene to be 0.028. The range of zoning in the pyroxenoid of 10058 varies from grain to grain but it seems to extend from compositions near $\text{Ca}_{22}\text{Mg}_{20}\text{Fe}_{58}$ to $\text{Ca}_{12}\text{Mg}_2\text{Fe}_{86}$. More lime-rich final compositions near $\text{Ca}_{20}\text{Mg}_2\text{Fe}_{78}$ are encountered in other grains. These iron-rich compositions are close to $\text{Ca}_{15}\text{Mg}_0\text{Fe}_{85}$ synthesized by LINDSLEY (1967). It should be noted that they lie well within the forbidden zone for low pressure pyroxene (and presumably pyroxenoid), as defined by LINDSLEY and MUNOZ (1969).

The analyses of the pyroxenoid reported on Table 4 call for little comment as they continue the fractionation trend for minor elements established for the pyroxene. The slight increase in titania between analyses 5 and 6 cannot be significant but the analyses do extend the field of the pyroxenoid to compositions richer in magnesium and poorer in lime than have hitherto been reported.

An explanation for the appearance of the pyroxenoid is not obvious; it has clearly taken the role in which fayalitic olivine is often found and its appearance must also be related to the failure of iron-enriched pigeonites to crystallize. Another factor of considerable importance is the subcalcic ferroaugite trend which extends and lies below the hatched trend indicated by LINDSLEY and MUNOZ (1969, Fig. 7) for this mineral as it defines a polythermal boundary of the two-pyroxene field for iron-rich, silica-saturated rocks.

Consideration of the textures of the pyroxenoid and the apparent lack of any compositional break or overlap with the pyroxene appears to indicate that at high temperatures in the absence of a stable fayalite field, a single clinopyroxene phase may have been able to crystallize almost as far as the position on the hedenbergite-ferrosilite join where LINDSLEY's synthetic pyroxenoid occurs. Subsequently the more iron-rich portion may have inverted to triclinic symmetry. Evidence obtained by LINDSLEY and BURNHAM (1970) indicates a triangular stability field for the pyroxenoid mineral ($\text{Ca}_{15}\text{Mg}_0\text{Fe}_{85}$) lying between invariant points located approximately at 9.5 kbar, 1190°C; 10 kbar, 1040°C; and 17.5 kbar, 1270°C. At lower pressures it breaks down to a lime-enriched pyroxenoid, fayalite and silica, while at lower temperatures it transforms to a clinopyroxene. In this case the association with tridymite and cristobalite must indicate a low pressure environment and hence a topochemical genesis may be invoked by inversion from a clearly unstable pyroxene. At a late stage the pyroxenoid itself seems to be breaking down to fayalite and a silica mineral. Fayalite and hedenbergite are the last ferromagnesian minerals to crystallize in the silica-enriched residuum just as in the granophyres of the Skaergaard intrusion.

In Fig. 5, the pyroxene-pyroxenoid trend of 10058 is compared with that of the Skaergaard intrusion, in which the pigeonite trend is defined from MUIR's unpublished data, optical and probe, on the late pigeonites. It may be significant that this trend terminates at the same composition as that of PHILPOTTS' (1966) pigeonite. In the Skaergaard intrusion uninverted pigeonite continues to crystallize in very small amount long after the reappearance of olivine. It should be pointed out that this

extended period of crystallization of pigeonite is subsequent to the sudden increase in lime content of the co-existing ferroaugite, as shown by the tie lines indicated in Fig. 5. At present no explanation of this lunar clinopyroxene trend can be offered; it may be concerned with silica activity, but this does not quite correlate with the Skaergaard data now reported. On the other hand the very delayed appearance of fayalite does fit in with the observations made by Lindsley and Munoz.

(iv) *Amphibole*

A dark green amphibole crystal was extracted from a small vug in rock 10058. It was approximately 0.5 mm in length, about 0.2 mm in width and less than 0.1 mm thick; prism faces were developed. X-ray data and a chemical analysis are given by GAY *et al.* (1970). The optical properties are as follows.

$n_\alpha = 1.638$	$n_\gamma - n_\alpha = 0.007$	Pleochroism
$n_\beta = 1.642$	Optic axial plane $\perp$ (010)	$X = \text{green}$
$n_\lambda = 1.645$	$Z = [010]$; $X \wedge [001] = 20^\circ$	$Y = \text{yellowish green}$
all ± 0.002	$2V_x = 60^\circ$ (approx).	$Z = \text{pea green}$
		$Z > X > Y$

The mineral has very strong dispersion and fails to extinguish in most directions in the prismatic zone. $X \wedge [001] = 23^\circ$ ($\lambda = 489 \text{ nm}$) and 18° ($\lambda = 656 \text{ nm}$). In convergent white light very strong dispersion was observed, for the YZ isogyre developed strong colour fringes and completely disappeared as Y was set vertical. It was not possible to determine $2V$ directly, as the fragment used for the optical investigation disintegrated before the indirect method could be applied.

The chemical analysis reveals that this is a very low alumina soda amphibole belonging essentially to the eckermannite–arfvedsonite series but with some affinity with richterite; the ratio $100 \text{ Mg} : (\text{Mg} + \text{Fe} + \text{Mn}) = 71$. This ratio and the orientation of the optic axial plane classify it as a rare variety, magnesioarfvedsonite, and its optical properties, high dispersion and low birefringence are in keeping with this. The refractive indices, particularly n_γ , are a little low but the mineral almost certainly has negligible ferric iron. A comparison may be made with the eckermannite from a nepheline syenite of Norra Karr reported by SUNDIUS (1945), but his amphibole contains over 6 per cent of alumina and 8 per cent of ferric iron.

The high soda of the analysis and the postulated presence of water are significant in pointing to the enrichment of these components in the late fluids.

(v) *Plagioclase*

In all cases the plagioclase has proved to be exceptionally calcic, extremes of composition encountered lying in the range An_{95} – An_{59} , but An_{100} has been reported. All crystals are flattened on (010); twinning is mostly on the Albite, Carlsbad and Albite–Carlsbad laws; many crystals are built up of four components, and twinning with vicinal composition planes subtending angles of about $1\frac{1}{2}^\circ$ with (010) is quite common. Pericline twinning is quite frequent, the angle σ in a twin in 10058 being -6° . An Acline A twin was determined in 10017, and a Banat twin was observed

in 10058 enclosed in the pyroxenoid. Measurements of plagioclase crystals are proceeding, but preliminary results indicate a transitional structural state $An_{7.9}$ in crystals virtually unzoned except at the extreme margins in 10017. In 10058 the bulk of the plagioclase is $An_{8.9}$ zoned marginally to $An_{8.6}$, but the structural state has not been established. In 10045 a high structural state was found with the crystals zoned from $An_{7.0}$ to $An_{5.9}$, but a few microphenocrysts are considerably more calcic ($An_{9.4}$). In 10058 the late plagioclase intergrown with the cristobalite has a composition $An_{7.7}$; it is associated with little rectangular patches of an optically-negative potash feldspar.

Analyses of plagioclases are given in Table 5. They include analyses from 10017 and 10045 which are in excellent agreement with the optical determinations reported above. As expected, the analyses are all poor in potash, but a considerable amount of FeO is present. The FeO reported (0.14%–0.55%) is thought to be integral to the feldspar, as inclusions were not present near where the microprobe analyses were made.

Table 5. Plagioclase analyses

	1	2	3	4	5	6	7	8	9
SiO ₂	44.21	44.90	45.72	47.34	49.57	45.35	45.6	47.29	47.57
TiO ₂	0.02	0.02	0.10	0.05	n.d.	n.d.	n.d.	n.d.	0.69
Al ₂ O ₃	35.68	34.78	34.53	33.36	32.11	34.81	34.5	33.41	22.57
FeO	0.14	0.21	0.30	0.58	0.44	0.55	0.44	0.26	6.10
MgO	0.10	0.05	0.07	0.07	0.11	0.08	0.06	0.10	8.51
CaO	19.28	18.92	18.64	18.06	15.71	18.38	18.58	16.72	14.14
Na ₂ O	0.46	0.60	0.70	1.17	1.89	0.88	0.89	2.04	0.70
K ₂ O	0.07	0.08	0.01	0.02	0.24	0.05	0.01	0.09	0.03
Totals	99.96	99.56	100.07	100.65	100.07	100.10	100.08	99.91	100.54
Number of ions on basis of 0 = 32									CIPW norm
Si	8.180	8.330	8.425	8.665	9.136	8.361	8.407	8.699	Or 0.18
Al	7.781	7.606	7.500	7.197	6.758	7.565	7.498	7.244	Ab 5.89
Ti	0.003	0.003	0.014	0.007	—	—	—	—	An 58.04
Fe	0.022	0.033	0.046	0.089	0.61	0.076	0.061	0.040	Di 9.49
Mg	0.028	0.014	0.019	0.019	0.30	0.022	0.016	0.027	Hy 20.50
Ca	3.822	3.761	3.680	3.542	3.103	3.631	3.671	3.295	Ol 4.46
Na	0.165	0.216	0.250	0.415	0.675	0.315	0.318	0.728	Cm 0.15
K	0.017	0.019	0.002	0.005	0.056	0.012	0.02	0.021	Ilm 1.30
Atomic %									
Ca	95.5	94.1	93.6	89.4	80.9	91.7	92.0	81.5	
Na	4.1	5.4	6.4	10.5	17.6	8.0	8.0	18.0	
K	0.4	0.5	—	0.1	1.5	—	—	0.5	

1. 10085-4-10a. Microphenocryst in microanorthositic fragment in fines. Mean of three analyses with Ca 94.6–96.1 at.%, Na 5.0–3.5 at. %.

2. 10085-4-10a. Pale green glassy fragment of maskelynite.

3. 10045-35-1. Unzoned larger crystal in basaltic rock.

4. 10045-35-1. Small crystal in basaltic rock.

5. 10017-12. Crystal in medium grained basaltic rock.

6. 10085-4-2. Crystal in doleritic chip in fines.

7. 10085-4-8. Crystal in doleritic chip in fines.

8. 10065-15-1. Recrystallized shocked plagioclase in microcrystalline partially devitrified glass fragment in microbreccia.

9. 10065-15-1. Microcrystalline partially devitrified glass enclosing plagioclase, analysis 8 this table. Analysis also includes MnO 0.13, Cr₂O₃ 0.10.

Table 6. Microprobe analyses of

	1	2	3	4	5	6	7	8
SiO ₂	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	0.14	n.d.
TiO ₂	53.73	53.80	55.43	54.18	53.50	59.42	53.95	56.30
Ti ₂ O ₃								
Al ₂ O ₃	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	1.64
Cr ₂ O ₃	0.75	0.92	1.15	1.15	0.46	0.45	0.66	0.34
FeO	43.03	41.02	38.80	38.83	44.49	35.43	43.80	32.39
MnO	0.39	0.38	0.36	0.37	0.42	0.28	0.40	0.34
MgO	1.90	3.20	3.71	4.90	0.96	4.14	0.97	9.63
CaO	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	0.08	0.44
Totals	99.80	99.32	99.45	99.43	99.83	99.72	100.00	101.08
Number of ions on basis 0 = 6								
Si	—	—	—	—	—	—	0.007	—
Ti ⁴⁺	—	—	—	—	—	—	—	—
Ti ³⁺	2.006	2.000	2.034	1.988	2.001	2.108	2.017	1.951
Cr	0.003	0.036	0.044	0.044	0.002	0.002	0.026	0.012
Al	—	—	—	—	—	—	—	0.089
Fe	1.786	1.696	1.582	1.582	1.860	1.512	1.821	1.248
Mn	0.016	0.016	0.014	0.016	0.018	0.012	0.017	0.013
Mg	0.140	0.236	0.270	0.356	0.072	0.294	0.072	0.661
Ca	—	—	—	—	—	—	0.004	0.022

1–6. 10045-35. Single point analyses on separate crystals of ilmenite in basaltic rock.

7. 10017-12. Average of 9 analyses of ilmenite in coarse basaltic rock.

8. 10085-4-10. Geikielo-ilmenite from very fine-grained anorthositic rock. Fragment in fines.

9. 10085-4-14. Magnesian ilmenite. Early separated crystal adjacent to spinel (analysis 10).

10. 10085-4-14 (a, b, c). Chrome-titano-spinel. Early separated crystal adjacent to magnesian ilmenite (analysis 9). Three analyses at separate points of a single crystal which shows slight incipient exsolution of ilmenite (?) in barely resolvable lamellae ($\sim 0.1 \mu\text{m}$). The structural formulae have been calculated on the basis of 0 = 32 with the titanium distributed between Ti³⁺ and Ti⁴⁺ so that total cations = 24.

11. 10085-4-10. Rutile associated with geikielo-ilmenite (analysis 8).

12. 10065-15-1a. Armalcolite microphenocryst in quenched oxide-rich olivine basalt, fragment in microbreccia.

The amount of FeO which is attributable to continuous fluorescence excited by neighbouring ilmenite crystals is likely to be less than 0.1%, as is indicated by the low TiO₂ in the analyses, which component would also be subject to this type of correction.

(vi) Opaque minerals

In the samples reported on in this paper the following have been observed: chrome-titano-spinel, ilmenite, rutile and ulvöspinel. The latter has not been confirmed by electron microprobe analysis.

(a) *Ilmenite*. This is a mineral that crystallizes from an early to almost the final stages of crystallization of the igneous rocks. The range of composition in the fine-grained basaltic rock 10045 is given in Table 6 analyses 1–6. These analyses, determined at single points, each in a separate crystal (without exsolution lamellae) show a wide range in composition (TiO₂ 53.50–59.42; Cr₂O₃ 0.45–1.15; FeO 35.43–44.49; MgO 0.96–4.90%). Alumina has not been determined, but the range reported from basaltic igneous rocks by several investigators is between 0.1 per cent and 0.36 per cent. (HAGGERTY *et al.*, 1970; SIMPSON and BOWIE, 1970; RAMDOHR and EL GORESY, 1970). The significant features of these analyses are the complementary variation of iron and

ilmenite, spinel, rutile, armalcolite

9	10a		10b		10c		11	12
0·20	0·40		0·40		0·40		n.d.	0·20
54·36	25·12	18·17	22·95	20·78	22·68	20·11	96·62	75·31
		6·28		1·95		2·31		
0·10	6·23		6·26		6·26		1·91	2·14
0·66	21·40		21·85		22·68		0·30	2·43
38·07	40·71		43·38		42·81		0·22	12·75
0·32	0·34		0·34		0·34		0·03	0·11
6·46	5·72		5·77		5·76		0·05	7·89
n.d.	0·01		0·01		0·01		0·37	0·14
100·27	99·93	99·26	100·96	100·74	100·94	100·68	99·50	100·97
			on basis 0 = 32			on basis 0 = 6		on basis 0 = 5
0·010	0·111	3·904	0·110	4·397	0·110	4·261	—	0·007
1·963	3·793		4·287		4·151			1·980
	1·457	8·193	0·447	7·210	0·529	7·475	2·912	
0·025	4·697		4·739		4·921			0·010
0·014	2·039	11·902	2·024	12·393	2·025	12·265	0·090	0·067
1·532	9·452		9·952		9·827			0·088
0·014	0·080	23·999	0·079	24·000	0·079	24·001	0·007	0·373
0·461	2·367		2·359		2·356			0·001
0·005	0·003		0·003		0·003		0·003	0·003
							0·016	0·411
								0·005

magnesium, the high chromium content, and the variable amount of titanium. There is an indication of excess titanium if calculated as Ti^{4+} , particularly in analysis 7 of Table 6, and some of the titanium may exist as Ti^{3+} .

Individual ilmenite crystals in rock 10045 must vary in composition and possibly be zoned in a complex manner, like the co-existing pyroxene crystals (Table 3, analyses 1–3 and Fig. 5). That early-separated ilmenite is richer in magnesia than that separating later has been confirmed in 10085-4-14₁₂ where larger and smaller idiomorphic crystals were analysed. This relationship is also suggested in 10085-4-14 (Table 6, analysis 10) where the ilmenite that has crystallized at the same time as the early-separating chrome–titano-spinel (Table 6, analysis 9) has 6·46% magnesia.

The highest content of magnesia (9·63%) is in a geikielo–ilmenite in a micro-anorthositic fragment (Table 6, analysis 8). There it is closely associated with rutile (Table 6, analysis 11) in a manner that suggests simultaneous crystallization rather than exsolution.

(b) *Chrome–titano-spinel*. This occurs in a medium-grained basaltic rock (10085-4-14) and is associated with magnesian ilmenite in a manner described above. Three analyses (Table 6, analyses 10, a, b, c) have been made at separate points within a single crystal, and a variation in composition is established. The analyses all show excess TiO_2 and a deficiency of other cations when the structural formula is calculated on the assumption that all titanium is Ti^{4+} . This suggests that some of the titanium exists as Ti^{3+} . The analyses have been recalculated by distributing the titanium between Ti^{4+} and Ti^{3+} in such a way that for oxygen = 32 the total of cations is 24. The calculated percentage weight of TiO_2 and Ti_2O_3 is indicated in the table as well as the appropriate new summation of the analyses. A range of Ti_2O_3 from 6·25 to 1·95 per cent is inferred, and it is to be noted that the higher Ti_2O_3 values are associated with the lower values for FeO and that each analysis has essentially the same

content of magnesia. The few minute exsolution lamellae (possibly ilmenite) are thought to have little effect on the estimation of the Ti_2O_3 content. Even were it significant, all the titanium must at the time of its crystallization have been included in the spinel.

(c) *Armalcolite*. An intermediate member of the series $\text{MgTi}_2\text{O}_5\text{--FeTi}_2\text{O}_5$, has been observed in one basaltic fragment from the fines and also as microphenocrysts in a quenched oxide-rich olivine basalt fragment in a microbreccia (10065-15-1a). Its composition is given in Table 6, analysis 12 and it is illustrated in Figs. 17 and 18. It occurs as tubular and skeletal crystals which may be replaced at the margins by magnesian ilmenite and minute highly reflecting areas which are probably rutile. A later generation of skeletal crystals of ilmenite occurs in the cryptocrystalline groundmass.

(d) *Buddeleyite* has been observed as a late crystallizing mineral in the groundmass of 10058-23. It is always closely associated with ilmenite either in composite grains or adjacent tabular crystals up to $75\text{ }\mu\text{m}$ in length. Compared with ilmenite it has a lower reflectivity and has a lilac tint.

(e) *Rutile*. Apart from the recorded occurrence in the microanorthositic fragment, rutile has only been provisionally identified as minute reddish-brown needles in the residuum of rock 10058.

(f) *Metallic iron*. Analyses have not been made of metallic iron that occurs in less than $3\text{ }\mu\text{m}$ drops enclosed in troilite in the igneous rocks. These are reported by several investigators including RAMDOHR and EL GORESY (1970). A trace of nickel and up to 0.7% cobalt are typical of these determinations. GAY *et al.* (this supplement) found one irregular fragment in the fines about 0.3 mm across associated with plagioclase and a chrome-titano-spinel showing marked narrow exsolution lamellae. The composition of the iron was Fe 99.2%, Ni 0.1%, Co 0.7%—a highly typical igneous value. The size and shape of this fragment and the absence of troilite suggest that a localized segregation of iron may have occurred.

Petrology

All the igneous analyses display a consanguinity in their characteristic enrichment of iron and titanium and in their very high chromium and very low sodium in relation to the iron-magnesia ratio. There are however a number of significant differences between analyses that call for comment. Firstly there is a limited variation in the iron-magnesia ratio which ranges from 67 to 76 in the published analyses and this must imply some degree of differentiation. Secondly there is a large range of values for alumina which varies from below 8 to over 13 per cent in different specimens of basaltic rocks. Titania too, although always abnormally high by terrestrial standards ranges from just over 7 to over 13 per cent in different specimens. From these observations it must be concluded either that rocks from two or more quite different sources have been collected or that some movement between liquid and early formed crystals has occurred.

The analyses of the igneous rocks of Table 1 described here are plotted in Fig. 7 which is constructed on a normative basis. Here the igneous types cluster within or just beyond the plagioclase-poor side of the encircled area that defines the welding and

coating glasses. As a comparison with terrestrial types, the analyses of two basaltic rocks from the floor of the Atlantic ocean (MUIR and TILLEY, 1964, 1966) are given. One of these is a cumulative type. From this diagram it can be seen that although some concentration of ilmenite has occurred in the lunar rocks there appears to be no great concentration of olivine or of clinopyroxene. On the other hand compared with terrestrial types that are believed to represent former liquids, all the lunar analyses represent rocks of a highly mafic character. The analyses however do suggest some variation in the amount of plagioclase.

In view of the anhydrous character and very low silica contents of the rocks there must be a very close correspondence between the normative and modal minerals.

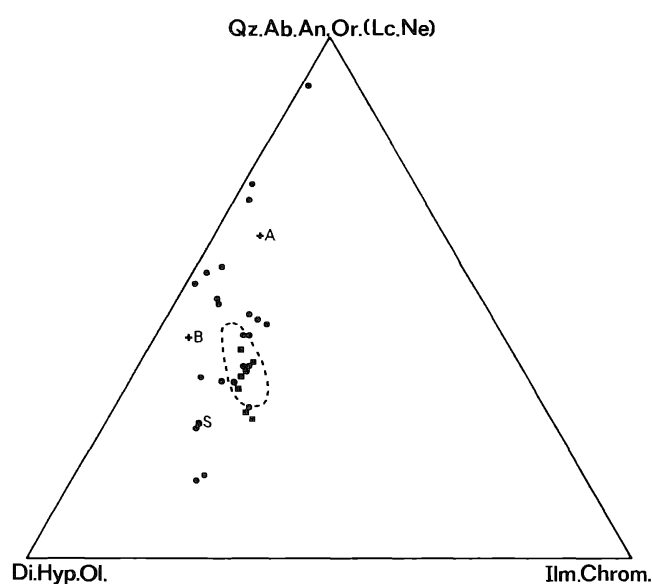


Fig. 7. Lunar glasses plotted in terms of normative components.

- Glassy spheres.
- S Glassy sphere, half of which analysed by SMALES (1970).
- Area within which welding and coating glasses fall (7 analyses).
- +A Olivine tholeiite, Mid-Atlantic Ridge. (MUIR and TILLEY, 1966, Table 3, No. 2).
- +B Olivine and pyroxene-enriched dolerite, Mid-Atlantic Ridge (MUIR and TILLEY, 1964b, Table 1, No. 1).

It is remarkable and significant therefore that all analyses so far calculated reveal the presence of small amounts of normative quartz; silica minerals or silica-rich glasses are prominent in the residuum. The normative pyroxenes plotted in Fig. 5 indicate well the stage of differentiation in their respective rocks and must bear a close relation to the mean compositions of the pyroxenes and pyroxenoids therein. This implies not only that the bulk composition of these phases is lime-poor as in tholeiitic rocks but also that it is relatively magnesian.

In view of the reduced importance of network silicates and the abundance of titanium in these magmas the viscosity should have been quite low and this might partly offset the reduced effect of gravity at an early stage of crystallization enabling some

movement between crystals and liquid to occur. On the other hand as the silica content of the residual liquid increased so gravitative separation of crystals and liquid would become more difficult; this might account for the features indicating extreme fractionation *in situ* that have been observed in the coarser grained rocks. It is interesting to note that notwithstanding the very low pO_2 inferred for these lunar magmas, the residual glass (Table 1, analysis 5), and the inferred composition of the residuum in the coarser rocks, indicate the development of a high-potash, high-silica fractionation trend.

MINERALOGY AND PETROLOGY: LUNAR FINES

This material may be considered as representative of the lunar soil with the addition of disintegrated fragments derived from the igneous rocks and microbreccia specimens which were included in the bulk sample container.

The principal components recognized are:

- A. Fragments of "normal" igneous rocks (or their constituent minerals) of the mare region.
- B. Fragments of microbreccia and their constituent rocks and minerals.
- C. Fragments of fine-grained anorthite-rich rocks (these are not represented among the normal igneous rocks).
- D. Fragments appearing like microagglutinate, made up of millimetre-sized soil or microbreccia fragments welded together and coated with glass.
- E. Relatively homogeneous glass fragments, and thick glass coatings.
- F. Glassy spheres from 3 mm to $\sim 1 \mu\text{m}$ in size.
- G. Meteoritic fragments.

A. Igneous rocks

The principal component consists of small angular fragments of igneous rock, angular fragments of crystals varying in size from 3 mm down to fragments less than a micron in dia. In the finest fraction (below $\sim 5 \mu\text{m}$) the ratio of crystalline fragments to glassy fragments decreases sharply.

The textural relations of the igneous rock fragments embrace relations normal to more slowly cooled basaltic rock (microgabbros and dolerites) as well as the variable textures found in quenched basaltic rocks. Porphyritic types are absent and although olivine, ilmenite and chrome-titano-spinel (Table 6, analyses 9 and 10) may occur as idiomorphic crystals, the textural relations are in conformity with simultaneous separation of ilmenite, pyroxene and plagioclase over the greater part of the crystallization history of the rocks. Their mineralogy (Table 3, analyses 4-9; Table 5, analyses 6-7), and by inference, their chemistry, does not differ significantly from the larger samples of igneous rocks that were collected in the same area except in so far as a proportion of the fragments show the effects of mild shock metamorphism. Although basaltic glasses are present, none are interpreted as quenched margins of basaltic flows or intrusions.

B. Microbreccias

The fragments of microbreccia show all the characters of this rock group and will be described under that heading. It should be appreciated that fragments of rocks and

minerals showing features attributable to shock metamorphism need not necessarily be derived from this group of rocks, but are also characteristic of the lunar soil itself.

C. *Anorthite-rich rocks*

A small proportion of the fines, estimated at less than 0.5 per cent is made up of dull white or grey fragments, about 2 mm across, which are distinctly lighter in colour than the normal basaltic igneous rocks. Identical rock types are found in the microbreccias.

The commonest type (10085-4-10) consists of microphenocrysts of plagioclase An 95.5 (Table 5, analysis 1) $\sim 250 \mu\text{m}$ in size, set in a granulitic mosaic (Fig. 12) of plagioclase, olivine, clinopyroxene $\sim 50 \mu\text{m}$ in dia., with approximately 5 per cent of geikielo-ilmenite (Table 6, analysis 8) and rutile (Table 6, analysis 11) and less than 0.1 per cent troilite with occasional small included crystals of metallic iron. No glass has been recognized in these microgranular or micropoikilitic types.

If the texture of these rocks is primary, which is unlikely but not impossible, they must represent the rapidly cooled products of an anorthositic or allivalitic magma. More probably they represent the products of pyrometamorphism of anorthositic or allivalitic rocks, the energy being supplied by meteoritic impact. It is also possible that some endogenous lunar explosion could have fragmented and fused an anorthositic cumulate to produce this rock type and distribute fragments into the area in which they are now found. The lower ilmenite and higher plagioclase content of these rocks distinguishes them from the normal igneous rocks of the Apollo 11 site. As the latter is in a Mare region these foreign anorthositic fragments might have come from a Highland region as has been suggested by a number of investigators (including WOOD *et al.*, 1970a). However, if feldsparphyric anorthite "basalts" or their shallow seated equivalents existed in another part of the Sea of Tranquillity, a cratering event there might provide petrographically indistinguishable material.

A second type, of dull white or pale green fragments, shows unambiguous evidence of shock metamorphism. Some of these fragments may be made up of the metamorphic maskelynite (Table 5, analysis 2) and show no evidence of flow or recrystallization. In others the texture of the original plagioclase-rich rock is seen on relict outlines of crystals which are now replaced by seriate blades of feldspar. Similar types have been observed in the microbreccias and in these interstitial or investing glass may be present. This preservation of the original texture of the anorthositic rock may indicate two periods of shock metamorphism in its history.

D. *Microagglutinate-like fragments*

Microagglutinate-like fragments (0.3–3.0 mm) make up a significant proportion ($\sim 10\%$) of the fines. They are the dominant component of the fines that can be extracted with a hand-magnet. Their characters are illustrated in Figs. 8–11. They consist of crystal fragments $< 250 \mu\text{m}$ in size from the lunar soil or minute fragments of microbreccia that are coated and welded together by a highly vesicular glass and are similar to those described by DUKE *et al.* (1970). The glass is pale brown, pale olive brown, or pale amber in thin section but in bulk the fragments are black.

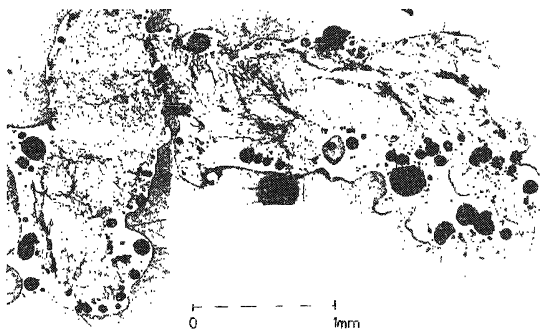


Fig. 8. 10085-14-17a. Fragments of lunar soil or microbreccia welded and fused together to give a composite group largely coated by vesicular glass. Reflected light, air.

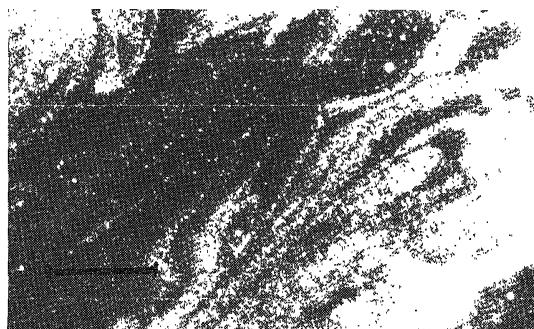


Fig. 9. 10085-4-17a. Coating glass viewed in reflected light showing complex flow pattern traced out by minute droplets of metallic iron. Scale bar 20 μm. Reflected light, oil.

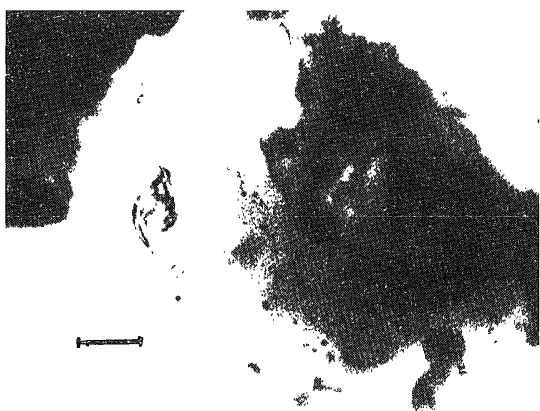


Fig. 10. 10085-4-17. Electron micrograph showing relict crystal fragments invested in glass. The spherical particles forming the fine trail in the glass are crystalline and more strongly absorbing than the glass itself; they are interpreted as being of metallic iron. Scale bar = 1000 Å.

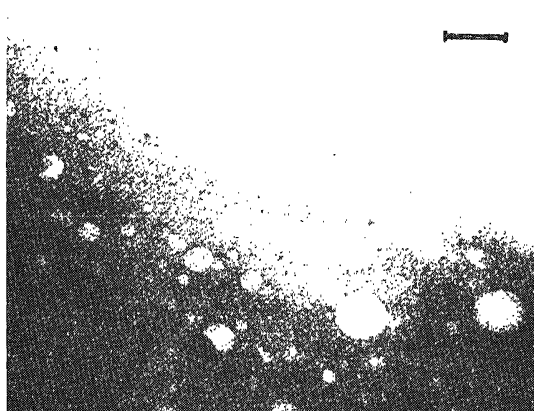


Fig. 11. 10085-4-17. Electron micrograph of lunar glass showing fine scale inhomogeneity. The lighter circular areas are spherical vesicles. The darker smaller specks are inclusions probably equivalent to the spherical particles seen in Fig. 10 above. Scale bar = 1000 Å.

The glass shows a complex flow pattern (Fig. 9) marked by slight differences in colour and by trails of immiscible droplets of iron, in the larger of which a rim of troilite may be detected. The maximum size of the droplets is of the order of 10 μm but in the transmission electron-microscope spherical particles down to 20 Å in dia. are visible (Fig. 10). It will be noticed that the spheres are relatively more absorbing than the glass itself, that the variation in contrast between different spheres of the same size is an electron diffraction contrast effect which establishes the crystallinity of the spheres. The detailed contrast in several of the larger droplets, seen at higher magnifications, establishes their monocrystalline character even though they are too small to show extinction fringes. The minutely vesicular nature of some of the glass is shown in Fig. 11 where the lighter circular areas represent spherical vesicles down to 20 Å in dia.

In total the metallic droplets probably amount to less than 0.2 per cent of the volume. The impression of greater abundance seen in the figures is due to all spheres within the depth of focus of the microscope being visible. The same effect can be well seen in overlapping vesicles in Fig. 11.

The composition of the larger, micron sized, spheres gave Fe $\sim$ 96% Ni $<$ 3%. ADLER *et al.* (1970) describes a sphere in frothy glass as having Ni tr, Co 0.4% similar to the composition of iron enclosed in troilite in the igneous rocks, and KEIL *et al.* (1970) reports Ni 0.5–19.8%, Co 0.35–0.95% but the data are for iron in glass as well as microbreccias. The ratio of metal to sulphide appears to be greater than in the igneous rocks and microbreccias. This suggests that sulphur vapour from the dissociation of troilite is in part responsible for the vesicular nature of the glass. The finely divided metal is probably responsible for the dark body colour of the glass which is consistently paler in thin section than that of glassy spheres of approximately the same composition. The darkening is analogous to the clouding of feldspars in certain thermal aureoles where unmetamorphosed feldspar is colourless or white in hand specimen and the metamorphosed feldspar almost black.

Table 7. Coating and welding glasses

	1	2	3
SiO ₂	40.81	41.77	41.08
TiO ₂	10.73	8.17	8.34
Al ₂ O ₃	10.20	12.31	13.07
Cr ₂ O ₃	0.32	0.35	0.32
FeO	19.51	15.94	17.65
MnO	0.23	0.22	0.23
MgO	6.11	9.53	8.11
CaO	11.92	11.06	9.93
Na ₂ O	0.34	0.36	0.34
K ₂ O	0.17	0.14	0.17
Totals	100.34	99.85	99.24
CIPW norm			
Q	2.41	—	—
Or	1.00	0.83	1.01
Ab	2.87	3.05	2.90
An	25.72	31.61	33.89
Di	27.66	19.06	13.00
Hy	19.57	20.79	32.49
Ol	—	2.61	0.27
Cm	0.47	0.52	0.47
Ilm	20.31	15.54	15.96

1. 10085-4-5. Brown tachylitic glass with occasional olivine and ilmenite microphenocrysts.

2. 10046-48-2. Olive-yellow glass seams in microbreccia, with complex flow pattern.

3. 10085-4-17a. Pale-brown vesicular glass with complex flow-pattern which coats and welds together fragments of lunar soil.

The origin of these glasses is puzzling; their composition approximates to that of microbreccia or lunar basalt (Table 7, analysis 3) and Fig. 7. They may be formed by some form of flash heating. Possibly this was induced by impact shock; this would explain the contorted flow pattern that is developed and relate these coating glasses to

the unique group of glassy spheres which show similar finely disseminated and flow-banded metal droplets; the glassy spheres being developed nearer the impact. On the other hand it is not conceivable that these fragments have travelled far from their source while the coating glass was in a fluid state, as the irregular liquid coating now present would be unstable and tend towards a rounded outline. Alternatively the melting may have been induced by the passage over the ground or above it, of an incandescent cloud of impact produced particles. Shock waves generated by the turbulence in such a cloud might be sufficient to induce a differential flow pattern in the coating glasses that its heat produced. This hypothesis has affinities with the base surge hypothesis of FISHER and WATERS (1969) considered by MCKAY (1970).

E. *Homogeneous glasses*

Relatively homogeneous glasses without a complex flow pattern have also been found as millimetre sized fragments in the fines.

Two were studied in some detail. 10085-4-5 is a brown-black glassy chip. Sporadic phenocrysts of spinel (Cr, Ti rich), ilmenite, olivine (Table 2, analysis 1) are set in glass (Table 7, analysis 1). It will be noted that the glass is quartz-normative and contains modal olivine. In Fig. 7 it plots within the field labelled coating glasses. No metallic iron or troilite were observed, but in a 2 mm × 2 mm fragment this is not too surprising. This glass could equally well have originated as a primary chilled edge to a basalt flow, as a chilled basaltic splash or have formed from complete shock melting of a basaltic rock.

The second example 10085-4-14₁₂ is a brown glass forming a coating to a basaltic rock. The latter contains two generations of idiomorphic tabular ilmenite crystals. These lie discordantly to the glass contact and are sharply cut off without any decrease in crystal size towards the contact. The glass again is without flow bands, is homogeneous and contains a few isolated immiscible iron-troilite drops. As there is such a sharp contact between glass and crystalline rock, with no evidence of partial melting, this fragment is probably derived from a basaltic splash (LSPET, 1969). Similar glasses in contact with soil aggregate or microbreccia fragments have been observed; here the contact is much more irregular (10085-4-14₉) and melting or incorporation of material has taken place, as ilmenite and pyroxene occur as rounded residuals in the glass, which also contains immiscible iron (Fe 96.1%, Ni 3.1%)—troilite drops.

F. *Glassy spheres*

Glassy spheres are a small component of the fines and range in diameter from 3 mm to $\sim 1 \mu\text{m}$. They are predominantly spherical but ellipsoidal and dumb-bell shaped forms occur, suggesting oscillation of drops during flight.

Colourless, pale green and pale straw coloured spheres are more common in the smaller size ranges, but amber, brown and black spheres occur in all sizes. Under low magnification the surfaces appear smooth, frosted (abraded?) or dusted with particulate matter. Under the microscope the latter are seen to be shocked or unshocked mineral fragments, oxides, silicates, native iron and troilite, all of which may have been picked up in flight while the sphere was still viscous.

The smaller spheres ($<200\text{ }\mu\text{m}$) are non-vesicular (on microscopic examination) and with increase in size a central vesicle or several smaller ones may be present. In the largest spheres they are common and examples have been seen where the vesicle comprises up to 50 per cent of the total volume of the sphere.

Polished mounts and thin sections were made of 20 separated spheres from the lunar fines, and others were examined in thin sections of the microbreccias.

One type of sphere is probably closely related in composition (and origin) to the coating glasses with the complex flow pattern described previously. They show a complex cellular or contorted flow pattern of metallic iron droplets with troilite which range in size from a few around $5\text{ }\mu\text{m}$ to many in the submicron range. The larger immiscible drops again are composite, with rounded drops of metallic iron (1–3% Ni) set in troilite: smaller metallic drops may have a film of troilite around a central iron core. These are similar to those described by DUKE *et al.* (1970). It is suspected that only metallic iron is present in the submicron sized drops. This type of glassy sphere is always vesicular, often highly so, and may contain resorbed ilmenite and clinopyroxene. If we relate the origin of this type of sphere to the comparable glassy coatings an impact theory would have them arising in an area of higher temperature or as spin-offs from liquid glass coatings in flight. If they arose by the passage overhead of an incandescent cloud, they might be produced when a turbulence locally swept the cloud across the ground lifting and carrying part of the lunar soil into the hotter parts of the cloud where the temperature was sufficiently high to fuse the lunar soil particles and so produce glassy drops.

The ratio of iron to troilite is much greater in all the glasses just described than it is in the igneous rocks, $\sim 8\text{ vol. \%}$ in troilite (SKINNER, 1970), more than 50% iron in drops in glass. The iron in troilite in igneous rocks though rounded in outline occasionally shows a crystal outline (more marked when plucked out by the polishing process), whereas in the glasses, the iron, irrespective of ultimate origin, is always as rounded drops where it occurs in troilite. Possibly this is due to the fact that on rapid heating the two phases did not homogenize and coexisted as two liquids for a short period. This might be expected on rapid heating to supraliquidus temperatures and would be frozen in on rapid cooling.

The majority of the glassy spheres are free from inclusions of pre-existing minerals other than a very small amount of metallic iron and troilite. Such crystalline products as they contain are either devitrification products or the products of normal high temperature crystallization. The former are more commonly plagioclase and clinopyroxene, the latter, spinel, olivine, ilmenite and possibly rutile and armalcolite (the last two have not been confirmed analytically).

The iron and troilite in composite drops may show a relative concentration at or near the margin of the spheres and is either material of meteoritic origin picked up in flight or, as suggested by RAMDOHR and EL GORESY (1970), segregated from within the sphere. The metal and troilite may occur as submicron droplets in the periphery of the glass in a similar manner to that described by EL GORESY (1968). Metal may also occur as occasional angular fragments sticking into the surface of the glass or as 10–25 μm -sized thick lens-like masses on the surface of the spheres. The latter are convex against the glass and the outer surface projects above the glass. These

relations are consistent with the metal troilite drops having been picked up in flight and being of meteoritic origin. The nickel content of the iron, both internal and peripheral, varies from sphere to sphere, in some cases it is almost absent, in others up to 13 per cent. These observations have been described by several investigators, among them RAMDOHR and EL GORESY (1970), DUKE *et al.* (1970) and KEIL *et al.* (1970). A real problem exists here in sorting out their endogenous and exogenous character. It is possible that nickel enrichment of the iron, by reaction of iron with silicate liquid may take place during an ultra high temperature stage (2000–5000°C) through which some spheres may have passed. Differential vapour loss might also occur. There is an indication of this process in the predominantly olivine-normative character of the glassy spheres (Table 8) as compared with the predominantly quartz-normative character of the Apollo 11 basaltic rocks. However the possibility that this reflects the original composition of the source region of the spheres must not be excluded.

One sphere, yellow brown in colour (10085-34-4) was cut in two and the glass analysed (Table 8, analysis 1), the other half was analysed for trace elements by SMALES *et al.* (1970). The glass encloses a large composite iron-troilite drop, the metal phase

Table 8. Analyses of glassy spheres

	1	2	3	4	5	6	7	8	9	10	11
SiO ₂	41.61	46.22	41.98	45.79	42.21	36.99	29.87	45.41	41.14	45.52	40.72
TiO ₂	7.21	0.81	6.91	1.18	8.47	7.78	7.32	1.87	11.38	3.43	6.63
Al ₂ O ₃	8.95	19.46	14.73	24.18	11.32	16.85	22.93	18.88	8.48	16.06	17.64
Cr ₂ O ₃	0.48	0.37	0.36	0.18	0.32	0.17	0.15	0.22	0.32	0.35	0.23
FeO	21.74	11.33	14.86	6.01	16.87	14.50	12.00	9.25	19.21	12.21	13.04
MnO	0.24	0.19	0.20	0.09	0.20	0.17	0.14	0.11	0.21	0.22	0.22
MgO	8.50	8.96	8.53	7.20	8.67	9.29	10.52	11.44	10.16	8.78	7.88
CaO	9.97	12.54	12.02	14.41	10.86	13.77	16.87	11.41	10.61	11.42	12.94
Na ₂ O	0.24	0.49	0.56	0.28	0.28	0.04	0.08	0.77	0.49	0.81	0.06
K ₂ O	0.11	0.14	0.06	0.06	0.21	0.08	0.08	0.32	0.45	0.20	0.08
Totals	99.05	100.51	100.21	100.01	99.41	99.64	99.94	99.68	99.46	99.00	99.44
CIPW norm											
Q	—	—	—	—	1.18	—	—	—	2.76	—	—
Or	0.66	0.82	0.35	0.35	1.25	—	—	1.90	2.72	1.19	0.48
Ab	2.05	4.13	4.73	2.37	2.38	—	—	6.54	4.17	6.92	0.51
An	23.24	50.23	37.42	66.26	29.18	45.72	—	47.27	19.74	40.00	47.90
Di	22.30	9.61	18.10	4.23	20.54	16.91	—	7.70	27.39	14.14	13.61
Hy	33.44	22.86	19.09	22.45	28.81	—	—	16.69	21.08	26.48	19.50
Ol	3.77	10.28	6.68	1.83	—	21.09	—	16.02	—	4.17	5.01
Cm	0.71	0.54	0.53	0.27	0.47	0.25	—	0.33	0.47	0.52	0.34
Ilm	13.81	1.53	13.10	2.24	16.18	14.38	—	3.56	21.73	6.58	12.66

1. 10085-30-4. Brown glass sphere from lunar fines, other half analysed by Smales.
 2. 10046-48-2. Devitrified glass sphere
 3. 10046-48-2. Orange coloured glass sphere
 4. 10046-48-2. Minute colourless glass sphere
 5. 10065-15-1. Broken orange yellow glass sphere
 6. 10065-15-1. Pale yellow glass sphere, included in norm Lc.37, Ne.18, Cs.64
 7. 10065-15-1. Very small pale yellow glass sphere—norm not calculable
 8. 10065-15-1. Orange brown glass fragment
 9. 10065-15-1. Lilac-tinted reddish sphere
 10. 10085-4-18. Pale brown glass sphere
 11. 10085-4-18. Darker brown glass sphere
- } from microbreccia.
- } from microbreccia.
- } from lunar fines.

has Fe 97.4%, Ni 0.6% (i.e. a near igneous composition), the sulphide contains Ni 0.16%. When plotted in terms of normative minerals, the glass (Fig. 7 ●S) is enriched in pyroxene and olivine compared with the Apollo 11 rocks.

Apart from the spheres with contorted flow-patterns of metallic iron droplets, the proportions of the different types as recognized by colour and by range of composition appear to be the same in the lunar fines as in the microbreccias. Representative analyses are quoted in Table 8, and plotted on a normative basis in Fig. 7. Here it is seen that the coating glasses cluster around the composition of the lunar basalt and microbreccia. The glassy spheres show a much wider spread of composition, though compositions with similar normative ratios to the basalts are common. One point previously mentioned is that in the samples analysed the ratio of olivine-normative to quartz-normative compositions is much higher than in the analysed rocks at the Apollo 11 site. This point has been noted among others by ADLER *et al.* (1970) and

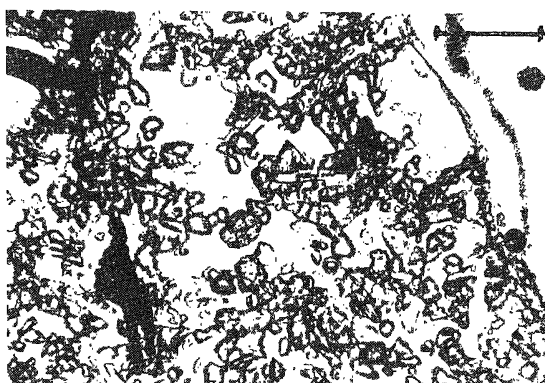


Fig. 12. 10085-4-10. Microanorthositic fragment. This is composed of geikielo-ilmenite (black) and small drop-like crystals of olivine and clinopyroxene, set in a base of tabular and interlocking crystals of plagioclase. Scale bar 100 μm .

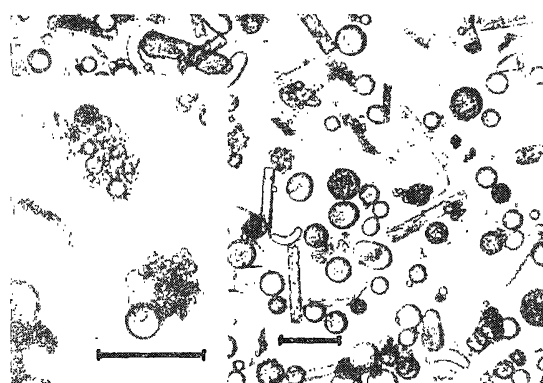


Fig. 13. Small glass spheres of basaltic composition extracted from a sample of Pele's Hair. The inset shows clusters of spheres down to 10 μm in size. Scale bar 100 μm .

FREDRIKSSON *et al.* (1970) and could reflect a source region with a different rock composition or a loss of SiO_2 , with or without other components, in the high temperature history of the spheres (WALTER, 1967). Though some spheres are low in alkalis compared with lunar basaltic rocks this is not necessarily due to loss of alkalis, for many spheres probably arose by impact fusion of a crystalline basaltic rock in which individual droplets reflect the original variation in mineralogy over small volumes. A study of the Or:Ab:An ratio in the norm of a larger number of analyses might establish alkali loss. This might have occurred in analysis 11, Table 8 but it equally well could have come from a rock with an original high anorthite content. Analyses 6 and 7 (Table 8) are definitely of abnormal composition showing low SiO_2 , comparable MgO contents and very high $\text{CaO}:\text{Na}_2\text{O} + \text{K}_2\text{O}$ ratios. The former has leucite and nepheline in the CIPW norm, and the CIPW norm of the latter is not calculable. These two at least must have modified compositions if derived from lunar basalt or they must come from some so far unrecognized lunar mineral assemblage.

Impact fusion of basalt rocks, with the exception previously noted, probably best explains the origin and composition of most glassy spheres. Alternatively those approximating to magmatic (basaltic) compositions may have arisen by a lava-fountaining process in the same way as the 250–5 μm spheres found associated with Pele's hair (Fig. 13) or any forced discharge of hot, dust-laden incandescent gas such as occurs in the furnaces of coal-burning power stations.

The surfaces of some of the glasses have been examined with the scanning electron microscope. Topographic features revealed are similar to those described by CARTER and MACGREGOR (1970) and include micro-impact craters. Some show a smooth crater in a central boss surrounded by a moat in which radial fractures may be marked and an outermost zone with shallow inwardly inclined concentric fractures. In Fig. 14 a fractured plate can be seen still attached to the surface. Some holes with very smooth rims (Fig. 15) in the surface of the glassy spheres show no fracture features and can probably be interpreted as arising from bursting gas bubbles; in some cases small domes of similar dimensions to the holes are also visible and are almost certainly subcutaneous bubbles. Alternatively, both the holes with re-entrant rims and those with raised rims, which have also been observed, might have been formed by impact of small particles during the time when the glass sphere was still a viscous fluid.

Circular features on the surface which are quite distinct appear as bright areas in a back-scattered electron image (Fig. 16) indicating the presence of particles with a mean atomic number appreciably greater than that of the silicate. One sphere was found to have approximately 20 per cent of the area of one part of its surface covered with particles of dia. $<0.5 \mu\text{m}$. Examination in the combined electron-microscope-microanalyser (DUNCUMB, 1966) indicated that the larger of these were iron-rich ($\text{Fe} > 70\%$) and that the Ti, Ni and S contents were almost certainly <5 per cent. This result suggests an identity with the nearly pure iron of the larger particles which are observable in sections of glass spheres (RAMDOHR and EL GORESY, 1970) but does not preclude the presence of particles with a nickel content corresponding to meteoritic iron (MCKAY *et al.*, 1970).

G. Meteoritic fragments

Modified and melted material, most probably of meteoritic origin, associated with lunar glasses has already been described in the previous section.

One normal meteorite fragment was extracted from the sample of fines. It is roughly equant in shape, about 2.5 mm in size and composed of polygonal crystals of kamacite, the largest of which was 0.5 mm across. The crystals are loosely attached and one was easily removed for X-ray examination. Two of the kamacite crystals have hemispherical embayments which may correspond to a position formerly occupied by troilite. The kamacite shows a marked development of Neumann bands and one of the crystals shows weak development of what may be the α - ϵ iron transformation. The composition of the kamacite is Fe 92.5%, Ni 6.5%, Co 0.5%. Minute (10–25 μm) oval drops of taenite are sparsely distributed in the kamacite and show zoning of nickel from ~ 9 per cent in the core to ~ 14 per cent in the rim.

Other evidence for meteoritic material is found in occasional angular fragments of kamacite and very rare taenite observed in the microbreccias. The only clue to

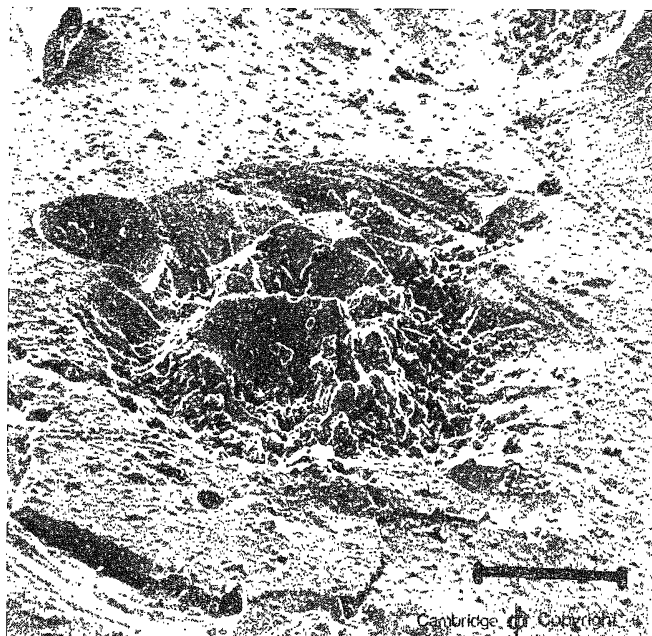


Fig. 14. 10085-30-7. Micro-impact crater on surface of a coating glass. (Scanning electron micrograph.) Scale bar = 50 μm .

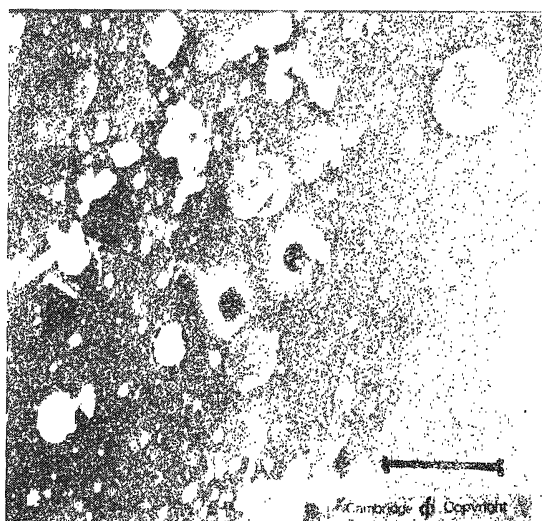


Fig. 15. 10085-4-29. Surface of glass sphere showing holes and raised domes. (Scanning electron micrograph.) Scale bar = 2.5 μm .

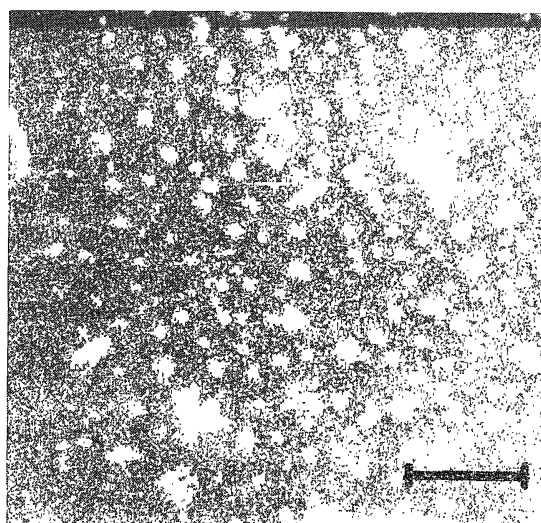


Fig. 16. 10085-4-27. Surface of glass sphere showing particles with mean atomic number higher than that of the matrix. (Scanning electron micrograph taken with back-scattered electrons.) Scale bar = 1 μm .

chondritic or achondritic material being present is the occasional rare taenite fragment reported from the microbreccia and in one crystalline sphere (10085-4-14), of which the silicate phases have not as yet been determined, in which taenite alone (or associated with troilite) contains from 9.5 to 17% nic. We may conclude that total contribution of meteoritic material to this part of the lunar surface appears to be small, less than 1 or 2 per cent.

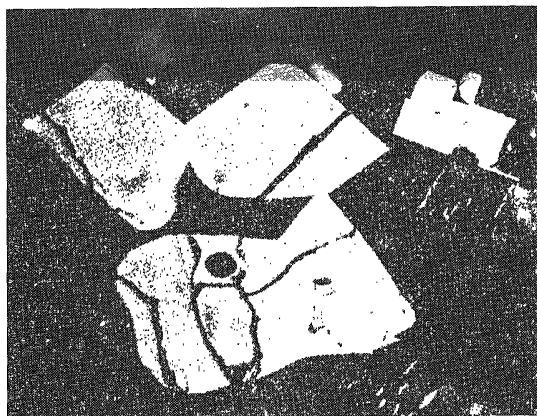


Fig. 17. 10065-15-1a. Idiomorphic microphenocrysts of armalcolite in oxide-rich olivine basalt from a fragment in microbreccia. Width of photograph 120 μm .

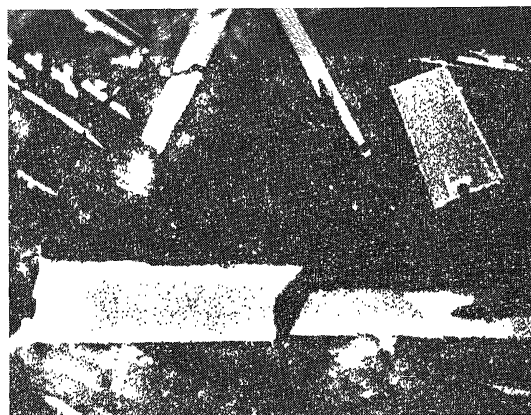


Fig. 18. 10065-15-1a. Idiomorphic microphenocrysts of armalcolite replaced marginally by ilmenite and rutile from an oxide-rich olivine basalt fragment in microbreccia. Width of photograph 120 μm .

MINERALOGY AND PETROLOGY: MICROBRECCIAS

Microbreccias 10046-48, 10048-42, 10060-28, 10065-15 were examined, specific attention being paid to the nature of their glassy components. The microbreccias contain angular fragments of basaltic igneous rocks and their component minerals, microanorthositic fragments, glassy spheres, seams and veins of glass, minute fragments of glass in the submicron size range, and a small proportion of meteoritic material.

The basaltic igneous fragments may show all the primary textures and range of grain size seen in the larger samples of these rocks and in addition a wide variety of quench textures. The compositions of the minerals do not differ significantly from those described earlier in this paper. The sole mineral observed only in the microbreccia was orthopyroxene in one microgranulitic fragment. A large proportion of the fragments of basaltic rock and individual mineral grains display a variety of cataclastic and shock effects. These commence with crystal granulation and fracturing of all minerals, at which stage twinning may be developed in ilmenite. Planar features in plagioclase and pyroxene were looked for but were not unambiguously observed. Partial and complete maskelynitization of plagioclase is seen both in rock fragments and in isolated plagioclase grains. In some cases the residual heat has been sufficient for maskelynite to crystallize, either to a seriate aggregate of small plagioclase laths or

small laths which show a preferred orientation within what were originally separate crystals. This feature is most clearly shown in almost monomineralic plagioclase aggregates and has been illustrated by WOOD *et al.* (1970b). Plagioclase glass in contorted and twisted fragments has also been observed and indicates that higher (fusion) temperatures have been attained. These may devitrify to give feathery swirled plagioclase aggregates where a high residual temperature was maintained.

Plagioclase-rich fragments are also found in which plagioclase is invested in a glass which may or may not have devitrified to a cryptocrystalline aggregate. The composition of the plagioclase and glass in one of these is quoted in Table 5, analyses 8 and 9. The anorthite content (81.5 at. %) is lower than most recorded in the microbreccias, and the norm of the glass corresponds to a gabbro in composition. Similar compositions have been demonstrated by ANDERSON *et al.* (1970).

Another plagioclase-rich type is occasionally observed in our microbreccia slides; these have also been described by ANDERSON *et al.* (1970) who appositely call them plagioclase vitrophyres. They show laths of anorthite-rich plagioclase with a sub-parallel orientation, up to 250 μm in length, set in almost colourless glass. These are the only feldspar-rich rock types to show a flow orientation and the only ones in which the texture is most reasonably interpreted as igneous.

In 10046-48-2 one unique angular fragment was observed composed of interlocking laths, $\sim 100 \mu\text{m}$ in length, of alkali feldspar with a silica mineral, probably tridymite, in the interstices. Electron microprobe analyses of the feldspar gave: SiO_2 64.63%, Al_2O_3 18.15%, FeO 0.10%, MgO 0.02%, CaO 0.33%, Na_2O 2.22%, K_2O 12.55%, total 98.00, which corresponds to K 77.5 at.%, Na 20.8 at.%, Ca 1.7 at.%. The presence of this fragment is what might be expected from a rock type related to the potassic interstitial liquid found in 10017-50 analysis of which is quoted (Table 1, analysis 5).

The microanorthositic fragments which also occur are similar in character to those already described from the lunar soil; they commonly have microgranular and microporphyritic textures. These can either be interpreted as igneous textures produced by rapid crystallization or as metamorphic textures. It is difficult to see in these fine grained rocks (300–50 μm) the evidence for significant primary cumulate processes which are implied by WOOD *et al.* (1970a).

Glassy spheres, occasionally broken, are conspicuous in all the microbreccias, and range from 0.5 mm down to submicron sizes. The very smallest, along with chips of glass and minerals, form a significant part of the material which bonds together the rock. The composition and colours are similar to those found in the soil, except that types with disseminated metallic iron are less common and two aberrant compositions have been observed (Table 8, analyses 6–7). Other analyses are quoted in Table 8 and plotted in Fig. 7. These have already been commented on in describing those from the lunar soil.

It is appropriate to make one or two additional comments at this juncture. WALTER and CLAYTON (1967) produced relative enrichment in O^{18} experimentally on heating and vapourizing acid glass and ONUMA *et al.* (1970) have shown that O^{18} is enriched in the spheres. This has also been demonstrated by EPSTEIN and TAYLOR (1970) who also showed relative enrichment of Si^{30} compared with lunar basaltic

rocks. This could be interpreted as confirming differential vapourization of components of the glassy spheres and could account for some of the abnormal compositions. The process would presumably operate at a maximum for those held longest at a high temperature or for those spheres which might have originated by condensation of vapour in the impact-produced cloud.

In considering the source of the glassy spheres, it is possible that some of them come from a Highland region. These should have compositions similar to the micro-anorthosite fragments, being poor in normative ilmenite and rich in normative anorthite, $\sim$ An 95 (Fig. 7). They may be difficult if not impossible to distinguish from spheres derived from plagioclase-rich volumes of the normal igneous rocks of the Mare regions.

Films and seams of vesicular glass are common in some microbreccia fragments. In the thin section 10046-48-2 such glass is pale olive yellow in colour and forms irregular veins 3–300 μ m in width. The glass is moulded on and encloses small fragments of microbreccia and a complex flow pattern is developed. The latter is marked by submicron-sized drops of what is interpreted as metallic iron; these are distributed in the same manner as those in the glass coating the microagglutinate fragments in the lunar soil. Such glass has probably formed in a short episode of intense heating which allowed some penetration (<5 mm) of the glass into the microbreccia. Some isolated fragments of similar glass occur in the microbreccia; these would imply more than one episode of intense heating and confirm the disruption and mixing of the microbreccia at some previous stage in its history.

Homogeneous and inhomogeneous pieces of different glasses also occur which are moulded on preexisting fragments within the microbreccias. These glasses represent material, unlike the glassy spheres, which was hot and plastic on impact with the ground. These would have formed at a stage when the present microbreccia was the unconsolidated lunar soil of which it is now a lithified representative.

Acknowledgments—This paper is the product of a collaborative investigation in which the prime responsibility for the various sections of the work undertaken was as follows: J.H.S. rock analyses; I.D.M. mineralogy and petrology of the igneous rocks; S.O.A. mineralogy and petrology of the lunar fines and microbreccias; J.V.P.L. direction of electron microprobe analysis and scanning electron microscopy; A.P. electron microprobe analysis; J.D.C. McC. electron microscopy. Considerable assistance was given by the crystallographic group headed by P. GAY.

We would in particular thank A. W. PORTER who spent much time and care in preparing thin sections for electron microprobe analysis, the Cambridge Scientific Instrument Company for access to their scanning electron microscope. Dr. P. DUNCUMB of the Tube Investments Research Laboratory for use of the electron-microscope-microanalyser and all the members of this department for their assistance and tolerance during the last four months.

REFERENCES

- ADLER I., WALTER L. S., LOWMAN P. D., GLASS B. P., FRENCH B. M., PHILPOTTS J. A., HEINRICH K. J. F. and GOLDSTEIN J. I. (1970) Electron microprobe analysis of lunar samples. *Science* **167**, 590–592.
- AGRELL S. O. (1970) The moon at Houston—Mineralogy. *Nature* **225**, 324–325.
- AGRELL S. O., SCOON J. H., MUIR I. D., LONG J. V. P., MCCONNELL J. D. C. and PECKETT A. (1970) Mineralogy and petrology of some lunar samples. *Science* **167**, 583–586.

Observations on the chemistry, mineralogy and petrology of some Apollo 11 lunar samples 127

- ANDERSON A. T., JR., CREWE A. V., GOLDSMITH J. R., MOORE P. B., NEWTON J. C., OLSEN E. J., SMITH J. V. and WYLLIE, P. J. (1970) Petrologic history of moon suggested by petrography, mineralogy, and crystallography. *Science* **167**, 587–590.
- ARRHENIUS G., ASUNMAA S., DREVER J. I., EVERSON J., FITZGERALD R. W., FRAZER J. Z., FUJITA H., HANOR J. S., LAL D., LIANG S. S., MACDOUGAL D., REID A. M., SINKANKAS J. and WILKENING L. (1970) Phase chemistry, structure, and radiation effects in lunar samples. *Science* **167**, 659–661.
- BROWN G. M., EMELEUS C. H., HOLLAND J. G. and PHILLIPS R. (1970) Petrographic, mineralogic, and X-ray fluorescence analysis of lunar igneous-type rocks and spherules. *Science* **167**, 599–601.
- CARTER J. L. and MACGREGOR I. D. (1970) Mineralogy, petrology, and surface features of lunar samples 10062,35, 10067,9, 10069,30, and 10085,16. *Science* **167**, 661–663.
- CHAO E. C. T., JAMES O. B., MINKIN J. A., BOREMAN J. A., JACKSON E. D. and RALEIGH C. B. (1970) Petrology of unshocked crystalline rocks and shock effects in lunar rocks and minerals. *Science* **167**, 644–647.
- DUKE M. B., WOO C. C., BIRD M. L., SELLERS G. A. and FINKELMAN R. B. (1970) Lunar soil: Size distribution and mineralogical constituents. *Science* **167**, 648–650.
- DUNCUMB P. (1966) *The Electron Microprobe*, pp. 490–499. John Wiley.
- EPSTEIN S. and TAYLOR H. P., JR. (1970) $^{18}\text{O}/^{16}\text{O}$, $^{30}\text{Si}/^{28}\text{Si}$, D/H, and $^{13}\text{C}/^{12}\text{C}$ studies of lunar rocks and minerals. *Science* **167**, 533–535.
- FISHER R. V. and WATERS A. C. (1969) Bed forms in base-surge deposits: Lunar implications. *Science* **165**, 1349–1352.
- FREDRIKSSON K., NELEN J., MELSON W. G., HENDERSON E. P. and ANDERSON C. A. (1970) Lunar glasses and microbreccias: Properties and origin. *Science* **167**, 664–666.
- GAY P., BANCROFT G. M. and BOWN M. G. (1970) Diffraction and Mössbauer studies of minerals from lunar soils and rocks. *Science* **167**, 626–628.
- EL GORESY A. (1968) The opaque minerals in impactite glasses. In *Shock Metamorphism of Natural Materials*, (editors B. M. French and N. M. Short), pp. 531–553. Mono.
- HAGGERTY S. E., BOYD F. R., BELL P. M., FINGER L. W. and BRYAN W. B. (1970) Iron–titanium oxides and olivine from 10020 and 10071. *Science* **167**, 613–615.
- JOEL N. and MUIR I. D. (1958) New techniques for the universal stage. I. An extinction curve method for the determination of the optical indicatrix. *Mineral. Mag.* **31**, 860–877.
- KEIL K., PRINZ M. and BUNCH T. E. (1970) Mineral chemistry of lunar samples. *Science* **167**, 597–599.
- LINDSLEY D. H. (1967) The join hedenbergite–ferrosilite at high pressures and temperatures. *Carnegie Inst. Wash. Yearb.* **65**, 230–234.
- LINDSLEY D. H. and BURNHAM C. P. (1970) In SMITH *et al.* Pyroxferroite, a new iron silicate from Tranquillity Base, moon. *Geochim. Cosmochim. Acta*, Supplement I.
- LINDSLEY D. H. and MUNOZ J. L. (1969) Subsolidus relations along the join hedenbergite–ferrosilite. *Amer. J. Sci.* **267A**, 295–324.
- LSPET (LUNAR SAMPLE PRELIMINARY EXAMINATION TEAM) (1969) Preliminary examination of lunar samples from Apollo 11. *Science* **165**, 1211–1227.
- MCKAY D. S., GREENWOOD W. R. and MORRISON D. A. (1970) Morphology and related chemistry of small lunar particles from Tranquillity Base. *Science* **167**, 654–656.
- MOSS A. A. and HEY M. H. (1961) Methods for the chemical analysis of meteorites. I. Siderites. *Mineral. Mag.* **32**, 802–816.
- MOSS A. A. and HEY M. H. (1967) Methods for the chemical analysis of meteorites. II. The major and some minor constituents of chondrites. *Mineral. Mag.* **36**, 101–119.
- MUIR I. D. and LONG J. V. P. (1965) Pyroxene relations in two Hawaiian hypersthene-bearing basalts. *Mineral. Mag.* **34**, 358–369.
- MUIR I. D. and TILLEY C. E. (1964) Iron-enrichment and pyroxene fractionation in tholeiites. *Geol. J.* **4**, 143–156.
- MUIR I. D. and TILLEY C. E. (1964) Basalts from the northern part of the rift zone of the Mid-Atlantic Ridge. *J. Petrol.* **5**, 409–434.
- MUIR I. D. and TILLEY C. E. (1966) Basalts from the northern part of the rift zone of the Mid-Atlantic Ridge. II. The Atlantis collections near 30°N. *J. Petrol.* **7**, 193–201.

128 S. O. AGRELL, J. H. SCOON, I. D. MUIR, J. V. P. LONG, J. D. C. McCONNELL and A. PECKETT

- ONUMA N., CLAYTON R. N. and MAYEDA T. K. (1970) Oxygen isotope fractionation between minerals and an estimate of the temperature of formation. *Science* **167**, 536–538.
- PHILPOTTS A. R. (1966) Origin of the anorthosite mangerite rocks in Southern Quebec. *J. Petrol.* **7**, 1–64.
- RAMDOHR P. and EL GORESY A. (1970) Opaque minerals of the lunar rocks and dust from Mare Tranquillitatis. *Science* **167**, 615–618.
- REED G. W., JR., JOVANOVIĆ S. and FUCHS L. H. (1970) Trace elements and accessory minerals in lunar samples. *Science* **167**, 501–503.
- ROSE H. J., CUTTITTA F., DWORNIK E. J., CARRON M. K., CHRISTIAN R. P., LINDSAY J. R., LIGON D. T. and LARSON R. R. (1970) Semimicro chemical and X-ray fluorescence analysis of lunar samples. *Science* **167**, 520–521.
- SEN GUPTA J. G. (1968) Determination of fluorine in silicate and phosphate rocks, micas and stony meteorites. *Anal. Chim. Acta* **42**, 119–125.
- SIMPSON P. R. and BOWIE S. H. U. (1970) Quantitative optical and electron-probe studies of the opaque phases. *Science* **167**, 619–621.
- SKINNER B. J. (1970) High crystallization temperatures indicated for igneous rocks from Tranquillity Base. *Science* **167**, 652–654.
- SMALES A. A., MAPPER D., WEBB M. S. W., WEBSTER R. K. and WILSON J. D. (1970) Elemental composition of lunar surface material. *Science* **167**, 509–512.
- SUNDIUS N. (1945) The composition of eckermannite and its position in the amphibole group. *Sverig. Geol. Unders. Årsbok* **39**, 1–7.
- SWEATMAN T. R. and LONG J. V. P. (1969) Quantitative electron-probe microanalysis of rock-forming minerals. *J. Petrol.* **10**, 332–379.
- WALTER L. S. (1967) Tektite compositional trends and experimental vapor fractionation of silicates. *Geochim. Cosmochim. Acta* **31**, 2043–2063.
- WALTER L. S. and CLAYTON R. (1967) Oxygen isotopes: Experimental vapor fractionation and variations in tektites. *Science* **156**, 1357–1358.
- WIK H. B. and OJANPERA P. (1970) Chemical analyses of the lunar samples 10017, 10072, and 10084. *Science* **167**, 531–532.
- WILCOX R. E. (1959) Universal stage accessory for direct determination of the three principal indices of refraction. *Amer. Mineral.* **44**, 1064–1067.
- WOOD J. A., DICKEY J. S., JR., MARVIN U. B. and POWELL B. N. (1970a) Lunar anorthosites. *Science* **167**, 602–604.
- WOOD J. A., MARVIN U. B., POWELL B. N. and DICKEY J. S., JR. (1970b) Mineralogy and petrology of the Apollo 11 lunar sample. *Smithson. Astrophys. Observ. Spec. Rep.* **307**.